

IEEE Recommended Practice for Market-Based Electric Vehicle Grid Integration

IEEE Systems, Man, and Cybernetics Society

Developed by the
Standards Committee

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IEEE Recommended Practice for Market-Based Electric Vehicle Grid Integration

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Standards Committee
of the
IEEE Systems, Man, and Cybernetics Society

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IEEE SA Standards Board

Abstract: Providing a set of market-based recommended practices for two-way interaction (vehicle-to-grid, V2G) technologies between electric vehicles (EVs) and the grid to facilitate the efficient integration and utilization of EVs as a flexible resource in the grid is the objective of this recommended practice. The overall architecture of the V2G system is described and a hierarchical zonal optimization operation model based on V2G technology to achieve the optimal allocation of charging and discharging power associated with electric vehicles is proposed. The application of market mechanisms in V2G technology and the importance of real-time measurement technology in V2G market settlement are emphasized, and a series of policy recommendations to guide the development of V2G technology are also proposed by this document. Through policy guidance and market incentives, the deep integration and effective interaction between EVs and the power grid and the cleaner transformation of energy as well as the construction of an efficient smart grid are promoted by this recommended practice.

Keywords: charging pile, hierarchical zonal aggregation, IEEE 3334™, market driven, real-time measurement technology, V2G technology

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Introduction

This introduction is not part of IEEE Std 3334-2026, IEEE Recommended Practice for Market-based Electric Vehicle Grid Integration.

As the global energy crisis and environmental problems become more and more significant, the popularization and application of electric vehicles (EVs), as an important carrier of clean energy, has become a keyway in which to promote the transformation of energy and support the reduction of carbon emissions. The two-way interaction between EVs and the grid [vehicle-to-grid (V2G)] technology has lifted knowledge to a new level, which not only improves the flexibility and stability of the grid, but also promotes the efficient integration and utilization of EVs as flexible resources in the grid. The purpose of this Standard is to introduce the “Recommended Practices for Market-Based Electric Vehicle Grid Integration”, and it elaborates on the structure of the V2G system, the characteristics of charging piles and their equipment, tiering and zoning aggregation strategies, market-driven demand responses, V2G performance testing, and policy recommendations to promote the development of V2G.

The V2G system is a complex integrated system that involves the deep integration of electric vehicles, charging facilities, communication networks and power grids. To construct an efficient and safe V2G system, charging facilities therefore need to be efficient, reliable, safe and intelligently controlled to meet the complex demands of V2G technology. The communication network is to help ensure real-time information interoperability between EVs, charging stations and power grids, including the transmission of key information such as charging status, power grid demand, and tariff changes. The communication network simultaneously supports direct communication between EVs and the grid to realize real-time transmission of energy demand and supply data. As the energy source and end user of the V2G system, the stability and flexibility of the power grid are crucial to the operation of the V2G system.

This standard describes a V2G-based, layered aggregation technique designed to address scheduling challenges from large-scale electric vehicle integration. By applying trusted technical approaches and aggregating EV resources at various points, the technique aims to utilize their potential for energy management, grid stabilization, and supporting the transition to a smarter, cleaner energy system.

The market-driven V2G demand response approach is the key to realizing the commercial utilization of V2G technology. This standard describes the establishment of a reliable vehicle-grid interactive resource aggregation process to participate in demand-side management and market trading mechanisms to incentivize EV users to participate in their grid demand responses. Therefore, EV users are guided to charge during the low periods of electricity demand and discharge during the peak periods, to alleviate the pressure on the power grid.

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IEEE Recommended Practice for Market-Based Electric Vehicle Grid Integration

1. Overview

1.1 Scope

This recommended practice provides data communication techniques, artificial intelligence methods, and integration planning strategies for the information exchange, energy storage, and economical operation of electric vehicles for grid integration. The concept of hierarchical and integrated organizational structure is defined, and the economic models of electric vehicles (EVs) for grid integration are established with incentive strategies. This recommended practice is suitable for managing and controlling vehicle-grid integration systems with safe and economical operation.

1.2 Purpose

This recommended practice helps manage and control vehicle-grid integrated systems, which will provide significant economic benefits and market demand that will benefit vehicle owners, electricity market operators, power utilities, power trading companies, the national grid, distribution system operators (DSOs) and independent system operators (ISOs).

1.3 Word usage

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The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

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⁷ The use of *will* is deprecated and cannot be used when stating mandatory requirements; *will* is only used in statements of fact.

2. Normative references

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IEC 61850:2026SER, Communication networks and systems for power utility automation—ALL PARTS.⁸

IEEE Std 2030.5™-2023, IEEE Standard for Smart Energy Profile Application Protocol.^{9,10}

ISO/DIS 15118-2:2018–06, Road vehicles—Vehicle-to-Grid Communication Interface—Part 2: Network and application protocol requirements.¹¹

3. Definitions, acronyms, and abbreviations

3.1 Definitions

For the purposes of this document, the following terms *and definitions apply*. *The IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.¹²

dispatchable electric vehicles (EVs): Electric vehicles (EVs) that accept electricity price incentives determined by the EV aggregator. They are willing to participate in the energy market and be controlled by the EV aggregator charge or discharge at the expense of consuming their battery life.

undispatchable electric vehicles (EVs): Electric vehicles (EVs) that do not accept the electricity price incentives determined by the EV aggregator. They are unwilling to participate in the energy market at the current trading period; however, they can participate in the energy market at another trading period and be categorized to the dispatchable EVs if the electricity price incentives are accepted.

3.2 Acronyms and abbreviations

6G	sixth generation
ADMS	advanced distribution management systems
AVC	automatic voltage control
AI	artificial intelligence
BMS	battery management systems
CPP	critical peak pricing
DER	distributed energy resources
DoD	depth of discharge
DSO	distribution system operator

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¹² *IEEE Standards Dictionary Online* is available at: <http://dictionary.ieee.org>. An IEEE account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

EMS	energy management software
EV	electric vehicle
EVSE	electric vehicle supply equipment
FCR	frequency containment reserve
GDPR	General Data Protection Regulation
IoT	Internet of Things
MARL	multi-agent reinforcement learning
M2M	machine-to-machine
ML	machine learning
MPC	model predictive control
OPF	optimal power flow
P2P	peer-to-peer
PLC	power line communication
R&D	research and development
RTP	real-time market pricing
SMI	smart metering infrastructures
SoC	state of charge
SWPDT	simultaneous wireless power and data transfer
ToU	time of use
UI	user interface
V2B	vehicle-to-building
V2G	vehicle-to-grid
V2H	vehicle-to-home
V2X	vehicle-to-everything
VPP	virtual power plant
WAVE	wireless access in vehicular environments

4. Vehicle-to-grid (V2G) system structure

4.1 V2G framework

4.1.1 V2G definition

Vehicle-to-grid (V2G) refers to a system in which EVs interact with the power grid to provide bi-directional energy flow (Sami et al. [B71]¹³). In this scenario, EVs act as distributed energy resources (DERs) by not only drawing power from the grid to charge EVs but also supplying stored energy back to the grid during periods of high demand or grid instability (J. Yang et al. [B90]). V2G technology transforms EVs from mere energy consumers to active participants in grid stability and energy management (M. Ansari et al. [B6]). This system enables EVs to contribute to grid services such as peak shaving, frequency regulation, and demand response, enhancing overall grid reliability, reducing carbon emissions and supporting efficient renewable energy integration.

¹³ The numbers in brackets correspond to those of the bibliography in [Annex B](#).

4.1.2 Related subjects

4.1.2.1 Power systems

V2G technology provides a range of grid services essential for stabilizing the power network. These include the following:

- *Frequency regulation:* V2G-enabled EVs should rapidly respond to frequency deviations by adjusting their charge/discharge rates, helping to maintain grid frequencies within safe limits (H. Liu et al. [B45]). This capability is critical for integrating intermittent renewable energy sources, for example wind and solar, which should cause significant fluctuations in power supply.
- *Voltage support:* By injecting or absorbing reactive power, V2G systems should support voltage regulation, helping maintain voltage levels within prescribed limits, particularly in areas with weaker grid infrastructure (N. Poursafar et al. [B64]).
- *Peak load reduction:* During peak demand periods, EVs should discharge power back into the grid, thereby reducing the burden on central generation facilities and helping to prevent the need for expensive peaking power plants (İ. Şengör et al. [B76]). This is particularly beneficial in hot summer months when air conditioning usage should cause high electricity demand.
- *Load balancing:* By coordinating the charging and discharging of EVs, V2G should help balance the load across the grid, shifting demand away from peak hours and utilizing off-peak generation, and this normally reduces the overall cost of electricity (Weckx and Driesen [B86]).

4.1.2.2 Communication and control

Reliable and efficient communication is fundamental to achieving optimal V2G integration. The interaction between EVs, charging stations, and grid operators requires real-time data exchange and control, facilitated by the following:

- *Communication protocols:* Protocols such as in Mültin [B56] and Garofalaki et al. [B27] enable standardized communications between vehicles, chargers, and grid systems. Mültin [B56] defines the required high-level communication between EVs and charging stations, supporting plug-and-charge functionality and secure energy transfer. Garofalaki et al. [B27] facilitates communication between charging stations and management systems, helping to ensure interoperability across different manufacturers and service providers.
- *Control algorithms:* Effective V2G operation requires sophisticated control algorithms to manage charging schedules, grid power quality, and helping to ensure optimal participation in ancillary service markets. Predictive algorithms, based on load forecasting and user preferences, are employed to decide when and how much energy is to be discharged back to the grid without compromising the EV owner's mobility needs.
- *Aggregation:* EV aggregators play a key role through coordinating multiple EVs to form a virtual power plant (VPP) (D. Yang et al. [B89]). This aggregation allows small, individual contributions to be pooled together so that participation in electricity markets and the provision of grid services at scale is achieved.

4.1.2.3 Market mechanisms

V2G has the potential to participate in various energy markets. To make V2G economically viable, several market-based approaches need to be considered, such as the following:

- *Wholesale energy market:* Aggregated V2G systems should participate in the wholesale energy market by providing electricity during peak price periods, thus benefiting from higher market prices (Mousavi et al. [B55]).
- *Capacity market:* EVs should be used as a reliable capacity resource, where they are committed to being available during peak events or emergencies (Duggan [B21]). Participating in capacity markets provides another revenue stream for V2G participants.
- *Ancillary services market:* By providing fast response capabilities like frequency regulation, EVs should participate in ancillary service markets (Z. Wu et al. [B88]). The quick ramp-up and down abilities of EV batteries make them ideal for these markets, where small but rapid power adjustments are critical for maintaining grid stability.
- *Incentive mechanisms:* To encourage EV owners to participate in V2G, incentive mechanisms such as dynamic pricing, tax benefits, or direct monetary compensation are often used. These incentives help offset the cost of battery wear and compensate for energy provided back to the grid.

V2G adoption is heavily influenced by regulatory policies and market rules such as follows:

- *Interconnection standards:* Regulations define how V2G systems should connect to the grid. Standards shall help ensure that V2G systems do not adversely impact grid stability and shall comply with safety requirements. Standards like IEEE Std 1547™ [B34] provide guidelines for interconnecting distributed resources, including V2G (Basso and DeBlasio [B9]).
- *Roles and responsibilities:* Clearly defined roles for EV owners, aggregators, utilities, and other stakeholders are essential for smooth V2G operation. Responsibilities include energy management, compensation, grid support, and helping to ensure compliance with regulatory requirements.
- *Incentive policies:* Policies that provide subsidies for bidirectional chargers, tax rebates for EV owners participating in V2G, or favorable electricity tariffs during off-peak hours should greatly enhance V2G uptake.

4.1.2.4 Power electronics

Bidirectional chargers are a critical component of V2G, as they facilitate energy transfer in both directions. They are as follows:

- *AC versus DC chargers:* Bidirectional chargers should be either ac or dc. DC bidirectional chargers are more common in V2G applications as they provide better control over the energy flow and reduce conversion losses, thereby enhancing the efficiency of energy transactions.
- *Charger standards:* Standards (Mouli et al. [B54]) support bidirectional power transfer, enabling the seamless integration of EVs into V2G operations. The charging standard in Mouli et al. [B54], for example, has been a pioneer in enabling bidirectional charging, allowing for practical V2G implementations.
- *Charger placement:* The strategic placement of bidirectional chargers is essential to increase V2G benefits. Public charging stations, workplaces, and residential areas are key locations that facilitate both convenience for EV users and effective grid support.

4.1.2.5 Battery devices

Battery degradation is a significant concern for EV owners considering V2G participation. It includes the following:

- *Cycle life*: The charging and discharging cycles in V2G increase the number of full equivalent cycles the battery undergoes, and this should reduce its lifespan. It is crucial to develop management strategies to reduce this impact, such as only allowing discharging during periods of high grid demand.
- *Battery management systems (BMS)*: Modern BMS are designed to optimize charging and discharging processes to extend battery life. They should limit the depth of discharge and control the rate of charging/discharging to mitigate the effects of V2G operation.
- *Economic compensation*: To offset battery wear costs, V2G participants should receive compensation that covers the accelerated degradation, making the V2G model more financially attractive. Research into the chemistry of new batteries with higher cycle life is also ongoing to make V2G participation more feasible.

4.1.2.6 Cybersecurity and privacy

V2G systems involve sensitive data exchanges, including location, charging patterns, and energy transactions (S. Acharya et al. [B1]). Cybersecurity and privacy concerns as follows:

- *Data security*: Robust encryption techniques are necessary to help ensure confidentiality and integrity of data exchanged between EVs, charging infrastructure, and grid operators. Protecting this data from cyber-attacks is crucial to help prevent malicious actors from disrupting grid operations or compromising user privacy.
- *Access control*: Authentication and authorization mechanisms shall be in place to help ensure that only legitimate entities should control or access the V2G system. This helps prevent unauthorized actions, such as maliciously discharging an EV battery.
- *Privacy concerns*: V2G data should reveal sensitive information about an individual's location and habits. Privacy-preserving data aggregation techniques should help mitigate these concerns, allowing data to be shared without compromising individual user privacy.

4.1.3 System control architecture (hierarchical coordination)

4.1.3.1 General

The V2G system control architecture is designed to manage interactions between EVs, aggregators, distribution systems, and the overall power grid. A hierarchical coordination approach is adopted to help ensure optimal integration of V2G resources, enabling efficient energy management at different levels while maintaining grid stability and meeting EV owners' needs. The control structure consists of four primary levels: device-level control, aggregator-level control, distribution system operator (DSO) control, and transmission system operator (TSO) control.

4.1.3.2 Device-level control

At the device level, individual EVs and their bidirectional chargers are managed to help ensure safe and effective participation in V2G operations. Local control mechanisms, including BMS, regulate charging and discharging based on user preferences, battery health, and grid signals. This level helps ensure the fundamental objectives of battery safety, lifecycle optimization, and local responsiveness, while also preparing EVs to engage in higher-level grid services.

4.1.3.3 Aggregator-level control

The aggregator level coordinates multiple EVs, forming a VPP to enable collective participation in energy markets and grid services. Aggregators are responsible for the following:

- Gathering data from individual EVs.
- Forecasting available capacity and scheduling charging and discharging.
- Submitting bids to energy and ancillary service markets.
- Coordinating with DSOs to align V2G operations with local grid conditions.

Aggregators play a crucial role in bridging the gap between individual EV users and grid operators, helping to ensure that distributed EV energy resources are efficiently utilized while maintaining grid stability.

4.1.3.4 DSO control

At the DSO level, V2G resources are integrated into the local distribution network. DSOs continuously monitor grid conditions, including voltage stability, power flows, and distribution network constraints. In collaboration with aggregators, DSOs should do the following:

- Request power injections from EVs to support peak demand.
- Direct EVs to charge when excess renewable energy is available.
- Leverage V2G to provide voltage control and mitigate local congestion.

The implementation of advanced distribution management systems (ADMS) enables real-time coordination of V2G activities, enhancing grid reliability and distribution network resilience.

4.1.3.5 TSO control

At the TSO level, the focus is on system-wide stability and grid balancing. The TSO interacts with DSOs and aggregators to help ensure that V2G resources contribute to frequency regulation, reserve power, and ancillary services. The TSO provides real-time grid signals that guide DSOs and aggregators in deploying V2G flexibility effectively. This coordination is particularly critical in high-renewable energy scenarios, where sudden fluctuations in power supply and demand require rapid-response mechanisms such as V2G-enabled frequency control.

4.1.3.6 Communication infrastructure

A robust communication infrastructure is essential to enable real-time data exchange across the hierarchical V2G control architecture. Technologies such as 5G, fiber optics, and Internet of Things-based (IoT-based) networks help ensure the following:

- EVs receive dynamic grid signals for optimized charging and discharging.
- Aggregators manage real-time energy scheduling and grid participation.
- Grid operators should dynamically adjust system parameters to maintain stability and resilience.

4.1.4 System control modes (coordination center, centralized, distributed)

4.1.4.1 General

The V2G system should operate under different control modes, each offering distinct advantages based on the requirements of grid stability, efficiency, and scalability. The four main control modes are coordination center-based, centralized, distributed, and local control, and each of these plays a critical role in enabling effective V2G integration.

4.1.4.2 Centralized control

In this mode, a central coordination center is responsible for overseeing and managing all V2G activities. The coordination center collects real-time data from EVs, aggregators, DSOs, and TSOs and uses this information to make informed decisions on how to best optimize energy flow (M. Elkholy et al. [B22]). This control mode provides a comprehensive view of the entire system, allowing for system-wide optimization and helping to ensure that energy is distributed to where it is needed most. The coordination center should facilitate large-scale grid services, such as frequency regulation and load balancing, by coordinating V2G activities across multiple geographical regions. Centralized control involves a single authority, typically the grid operator or an aggregator, who makes decisions for all participating EVs. This approach encourages effective oversight and should achieve significant efficiencies through large-scale coordination. Centralized control is particularly useful for managing grid services, such as peak load reduction and frequency regulation, as it enables the operator to have a clear understanding of available resources that facilitates effective decision making that increase overall grid stability. Centralized control should however be challenging in terms of scalability and should generate undesirable communication bottlenecks, especially as the number of EVs participating grows in the future.

4.1.4.3 Distributed control

In a distributed control mode, decision-making is spread across multiple entities, such as aggregators or even individual EVs (M. Liu et al. [B46]). Each entity operates based on local information and should coordinate with neighboring entities to achieve broader objectives. Distributed control offers greater scalability compared with centralized control, as the reliance on a single authority is reduced and therefore allows for faster decision-making at the local level. This mode is well-suited for managing localized grid issues, such as voltage support and load balancing, and should enhance system resilience by reducing the impact of a single point of failure.

4.1.4.4 Local control

Local control focuses on individual EVs and their interactions with the immediate environment, such as a home, workplace, or local charging station (Richardson, Flynn, and Keane [B68]). In this mode, EVs make decisions based on local factors, such as user preferences, battery state, and local grid conditions. Local control is essential for helping to ensure that the needs of EV owners are met while still contributing to grid stability. For example, an EV should decide to discharge to support a local microgrid during a power outage or charge during periods of excess solar generation at a residential site. While local control provides the highest level of autonomy for EV owners, its impact on overall grid stability is limited compared with other control modes.

4.2 National standard and specific requirements

4.2.1 V2G national standard requirements

4.2.1.1 General

National standards for V2G systems provide the foundation for integrating EVs as distributed energy resources within the power grid. These standards address critical aspects, including interconnection protocols, communication frameworks, market participation guidelines, and regulatory compliance, helping to ensure that V2G systems operate safely, reliably, and efficiently. By establishing clear requirements, national standards foster interoperability, encourage market alignment, and support widespread V2G adoption. Below is a detailed outline of the primary national standard requirements for V2G systems:

4.2.1.2 Interconnection standards for grid services

Interconnection standards play a crucial role in helping to ensure the safe and efficient integration of V2G systems with electrical grids. These standards establish technical and operational requirements for safely managing the bidirectional flow of energy between EVs and the power network. The primary goal is to maintain the stability, reliability, and safety of the grid while enabling the participation of V2G-enabled EVs. These systems shall meet stringent conditions to help ensure that the discharge of energy from EVs into the grid does not adversely affect the grid's operation or the performance of other connected devices.

IEEE Std 1547 [B34] serves as the foundational guideline for integrating DERs, including EV, into the grid. It addresses essential operational aspects such as voltage regulation, frequency stability, and power quality, all of which are critical for the smooth and safe operation of V2G systems. IEEE Std 1547 [B34] describes the framework for helping to ensure that bi-directional energy flows between EVs and the grid remain within predefined limits, thereby helping to prevent disruptions that might compromise grid integrity (Agashe [B3]). Compliance with this standard is vital to mitigate potential risks associated with grid overloads, voltage fluctuations, or frequency instabilities caused by uncontrolled or uncoordinated energy exchanges.

For V2G systems specifically, the integration guidelines emphasize several key technical parameters. Voltage control is one of the primary considerations that helps ensure that the voltage levels in both the EVs and the grid remain within operational limits. The charge and discharge cycles of EVs should introduce voltage fluctuations, which shall be effectively managed to help prevent destabilization of the grid. Voltage regulation mechanisms, such as automatic voltage control (AVC) or reactive power support, shall be implemented to maintain voltages within an acceptable range, as specified by local grid authorities.

In addition, frequency regulation is an essential requirement for V2G systems. The discharge of energy from EVs should lead to frequency deviations, which shall be carefully controlled to help ensure grid stability. V2G systems should be capable of responding to frequency variations by adjusting the power flow in a manner that helps restore the grid's frequency to its nominal value. This should be achieved through dynamic, real-time frequency-responsive controls that allow V2G systems to participate in grid frequency balancing without introducing additional instability.

Furthermore, power quality standards shall be adhered to in V2G systems to help prevent any disturbances that could affect the overall functioning of the grid or other connected devices. This includes managing harmonic distortion, voltage transients, and other power quality issues that should arise during energy transfers between EVs and the grid. The power delivered by V2G systems should meet predefined limits for harmonic content, voltage fluctuations, and other power quality parameters, as outlined by local or international standards.

In addition to adherence to these general guidelines, V2G systems shall also comply with local or regional interconnection rules that are tailored to specific grid infrastructures and capabilities. These rules help ensure that V2G systems should operate effectively within the unique characteristics of different grid environments while helping to avoid any new risks such as overload conditions, power surges, or grid faults. By helping to ensure compliance with both global and regional standards, V2G-enabled EVs should integrate seamlessly into the grid, supporting the broader goal of energy efficiency, grid flexibility, and the transition toward a more sustainable energy system.

4.2.1.3 Market participation standards

National standards for market participation provide the framework under which V2G systems should participate in energy, capacity, and ancillary service markets. These standards define the technical and operational conditions necessary for V2G systems to engage in market-based transactions and contribute to grid services. They set clear requirements for V2G participants, helping to ensure that resources meet specified performance criteria such as response time, capacity, availability, and reliability. By establishing these guidelines, national standards foster the economic viability of V2G systems, helping to ensure that EVs should generate revenue while supporting grid stability.

For participation in energy markets, aggregated V2G resources are typically required to discharge energy during peak demand periods, providing additional power to the grid when it is most needed. To engage in these wholesale energy markets, V2G systems shall meet specific criteria for response rates and duration of energy delivery. For example, V2G resources should be required to discharge energy within a defined response time (e.g., within minutes of a signal from the grid operator) and sustain that discharge for a specified duration (e.g., several hours). These standards help ensure that V2G systems should contribute to grid balancing and assist in managing supply-demand mismatches during times of peak load.

In capacity markets, V2G systems should be called upon to provide reliable power resources during peak or emergency events. These markets typically require participants to demonstrate the ability to provide a defined amount of energy or capacity when called upon, often within a specified time frame. To participate, V2G systems shall prove their capacity to provide energy when required, helping to ensure they are available and able to discharge energy rapidly if the grid experiences a sudden surge in demand or other disruptions. Reliability and availability are key factors in helping to ensure that V2G systems should meet these requirements and these contribute to the resilience of the grid.

In ancillary service markets, V2G systems should offer quick-response services such as frequency regulation and voltage support, which are vital for maintaining grid stability and operational efficiency. These services require V2G systems to respond quickly and accurately to signals from grid operators, adjusting their energy output in real time to support grid operations. To participate in ancillary services, V2G systems shall meet stringent technical requirements, including response accuracy, speed, and consistency. For instance, in frequency regulation, V2G resources should be required to adjust their discharges or charge rates in real-time to help stabilize grid frequency, often within seconds of receiving a signal.

Further, national standards often require that V2G participants demonstrate reliability and predictability. Grid operators need to be confident that V2G resources shall perform as expected, without introducing additional fluctuations or instability into the grid. This means that V2G systems shall be able to deliver energy consistently, with predictable performance characteristics. This reliability is essential in helping to ensure that EVs should be counted on to provide grid services when needed, especially during periods of high demand or in response to unforeseen grid disturbances.

4.2.1.4 Environmental and emission standards

Environmental standards are essential in promoting the integration of low-emission technologies within V2G systems, aligning with broader renewable energy and carbon reduction goals. These standards aim to reduce the environmental impact of V2G operations, helping to ensure that the deployment of V2G technologies contributes positively to emissions reduction targets and supports the transition to a sustainable energy future. Through the establishment of specific environmental criteria, these guidelines foster the use of renewable energy sources and incentivize the development of greener V2G solutions.

One key aspect of environmental standards for V2G systems is the integration of renewable energy within V2G operations. Standards should provide guidelines to help ensure that V2G systems, particularly EVs, are powered by renewable energy sources such as solar, wind, or hydropower. By requiring or encouraging the use of clean energy to charge EVs, these guidelines help ensure that V2G systems contribute to overall emissions reduction efforts. For instance, when EVs are charged using renewable energy, their subsequent discharge into the grid should support decarbonization goals, as the energy supplied does not contribute to increased greenhouse gas emissions. This approach allows V2G systems to serve as a low-carbon grid resource, supporting grid stability while reducing their environmental footprint.

Additionally, some national and regional guidelines specifically promote low-carbon grid support programs, where V2G systems powered by renewable energy should participate in grid services such as frequency regulation or demand response. In these programs, the emissions from V2G activities—whether through charging or discharging—are offset by renewable energy generation, helping to ensure that V2G participation does not increase the carbon footprint of the grid. These programs should involve accounting mechanisms that

track the emissions savings generated by V2G systems, aligning their operations with broader environmental policies and carbon reduction strategies.

Incentive structures are often incorporated within environmental standards to further encourage the use of renewable-powered V2G systems. National guidelines should include financial incentives such as tax credits, subsidies, or grants for individuals or organizations that deploy renewable-powered EVs or contribute to the decarbonization of the grid. These incentives lower the economic barriers to adopting V2G technologies and renewable energy solutions, making it more accessible for users to participate in sustainable grid operations. For example, tax credits should be offered to EV owners who charge their vehicles using solar energy, or subsidies should be available for V2G systems that provide grid services using renewable power. By offering these incentives, governments and regulatory bodies should accelerate the transition to greener and more sustainable energy systems.

4.2.1.5 Incentive and subsidy policies

National policies play a critical role in promoting the adoption and widespread integration of V2G systems through providing incentive mechanisms to EV owners. These incentives aim to lower the financial barriers for participation, making V2G operations more economically attractive and fostering broader engagement from individuals and organizations. By encouraging EV owners to participate in grid services such as demand response, frequency regulation, and energy storage, national policies shall help establish EVs as valuable, flexible, and reliable resources for grid management.

One of the primary incentive mechanisms in national policies is the provision of financial incentives. These should include tax credits or rebates for the purchase of bidirectional chargers, which enable the charging and discharging of energy between EVs and the grid. By reducing the upfront cost of purchasing and installing V2G-compatible chargers, these financial incentives make it easier for EV owners to transition to V2G-enabled systems. Additionally, some policies introduce dynamic pricing models that offer higher compensations for energy discharged during peak demand periods. This incentivizes EV owners to discharge energy when the grid needs it most, providing valuable support for grid stability and enabling EVs to generate additional revenue. In turn, this dynamic pricing should help offset the cost of EV ownership and operation, making V2G participation financially rewarding.

National policies should also introduce reduced electricity tariffs for EV owners who participate in V2G during off-peak hours. These tariffs incentivize EV owners to charge their vehicles when demand is lower and discharge energy during peak times, helping to balance supply and demand on the grid. By encouraging EV owners to utilize off-peak charging rates, these policies support grid efficiency and reduce the strain on the grid during periods of high consumption. These pricing models should significantly lower the cost of operating a V2G-enabled EV, further encouraging engagement from a wider range of consumers.

Another key concern in V2G participation is the potential impact on battery life due to additional charging and discharging cycles. The increased battery wear caused by V2G operations should raise concerns about the long-term cost-effectiveness of V2G from EV owners, as frequent cycling should reduce the lifespan of the battery. To address this, some national policies include compensation models to offset the additional wear and tear on EV batteries. These compensation schemes should offer financial remuneration or subsidies to cover the cost of battery replacement or maintenance, helping to ensure that the economic benefits of participating in V2G outweigh the costs of battery degradation. By mitigating the financial impact of battery wear, these policies encourage EV owners to participate in V2G programs without the concern of significant long-term costs.

4.2.1.6 Safety and compliance standards

Safety standards are crucial to help ensure that V2G systems operate without endangering users, the grid, or other electrical infrastructure. As V2G systems involve bidirectional energy flow between EVs and the grid, safety considerations shall be rigorously addressed at all levels of operation, from the charging infrastructure to the communication protocols. National safety requirements provide the necessary frameworks to mitigate potential risks associated with V2G technologies and help ensure that these systems comply with fundamental safety criteria.

One of the primary safety considerations in V2G systems is the protection of bidirectional chargers and other related equipment. National safety standards, such as UL 1741 (for grid-tied inverters; Hudson et al. [B31]) and IEC 61850:2026 SER (for EV charging systems; Di Giorgio et al. [B20]), set out specific safety protocols to protect users and help prevent damage to equipment.¹⁴ These protocols cover various aspects of electrical safety, including insulation, grounding, and overload protection. Insulation standards help ensure that the electrical components of the chargers are safely isolated from user-accessible parts, helping to prevent electrical shocks. Grounding protocols help ensure that any electrical faults are safely directed to the ground, reducing the risk of electric shock or fire. Overload protection helps ensure that chargers and related systems automatically disconnect or limit power in the event of excessive load or system faults, thereby helping to prevent damage to the equipment or the grid.

Additionally, compliance with these standards is essential to help ensure that V2G equipment operates reliably and safely within the broader electrical network. UL 1741 and similar standards establish requirements for inverters, which play a crucial role in managing the flow of electricity between EVs and the grid. These standards define the necessary safety tests, performance benchmarks, and operational safeguards for inverters, helping to ensure that they meet high-quality manufacturing and functional safety criteria before they are deployed in V2G systems.

Another critical aspect of safety standards for V2G systems is the development of guidelines that outline safe and effective emergency responses. Safety standards stipulate that V2G systems shall be able to respond quickly and safely to potential faults or grid disturbances. For example, automatic disconnection mechanisms are often required to help ensure that a V2G system immediately disconnects from the grid in the event of a grid fault or other destructive event caused by abnormal conditions. This emergency response feature helps prevent further damage to the equipment, protects the safety of users, and helps ensure that the grid remains stable during faults. Such mechanisms should also help mitigate the risk of electric shock to users and help prevent batteries from overcharging or deep discharging, which could damage the vehicle and/or other equipment.

In addition to these technical safeguards, safety standards should also reinforce user safety through proper labeling, user instructions, and the provision of safety features such as fault detection systems and overcurrent protection. Helping to ensure that users are aware of proper handling procedures and potential hazards is essential for helping to prevent accidents during operation, maintenance, or installation of V2G systems.

4.2.2 V2G communication protocol standards

4.2.2.1 General

Effective communication protocols are crucial for the seamless integration of V2G systems into the power grid. These standards establish the framework for secure, reliable, and efficient data exchange, helping to ensure proper coordination among EVs, charging stations, aggregators, and grid operators. One of the primary objectives of these protocols is to enable smart charging and discharging, allowing for dynamic decision-making based on grid conditions, market requirements, and user preferences. A standardized communication interface is essential for facilitating these processes, helping to ensure interoperability across different systems and devices.

¹⁴ Information on references can be found in [Clause 2](#).

The ISO/DIS 15118-2:2018–06 communication standard is widely recommended for integrating EVs into power grids (Schmutzler, Wietfeld, and Andersen [B72]), as it provides a robust framework for EVs and electric vehicle supply equipment (EVSE) to communicate efficiently with power grids (Brown [B11]). The primary goal of the ISO/DIS 15118-2:2018–06 standard is to standardize the communication interface between the vehicle and the grid, thus allowing EV users to easily access charging equipment and related services throughout regions such as Europe. This standard supports smart charging functions, enabling dynamic control of the charging process either at the EV level or the FOs (flexibility operators) level, depending on the system's configuration. The charging process is facilitated by a series of message exchanges between the EV and the EVSE.

In addition to ISO/DIS 15118-2:2018–06, IEC 61850:2026 SER is recommended for communication between the EVSE and FOs (Mackiewicz [B48]), helping to ensure seamless integration within the broader energy management framework. This standard is particularly useful for DERs, providing a common communication platform for various grid elements. In some cases, the use of smart metering infrastructures (SMIs) is proposed to relay information regarding the EV's charging state, therefore utilizing multiple access control protocols to facilitate communication.

For V2G applications, where control over both the charging and discharging processes is critical, the SAE protocol (Scholer et al. [B74]), established in January 2013, allows for the coordination of distributed energy resources and EVs. This protocol is particularly valuable for enabling V2G capability, as it facilitates real-time control over the charge and discharge processes, helping to ensure that EVs should serve as flexible grid resources. Information exchange with the EV is further supported by IEC/TR 61850-90-8.

Additionally, communication between charging stations and system operators or aggregators should be enhanced using protocols like those described in Garofalaki et al. [B27] and Hecht, Figgenger, and Sauer [B29]. These protocols provide alternative solutions for the communication and coordination between charging stations and the broader grid infrastructure, thus improving flexibility and enabling the integration of V2G services into demand response and other grid management applications.

4.2.2.2 Interoperability and standardization

Interoperability and standardization are essential for successful deployment and widespread adoption of V2G systems, enabling seamless communication and operation across a variety of devices, manufacturers, and grid infrastructures. Standardized communication protocols, such as ISO 15118 (see Mültin [B56]) for EV-to-grid communication and that discussed in Garofalaki et al. [B27] for communication between charging stations and backend systems, provide a consistent framework for data exchange, control, and coordination. These protocols define common data formats, messaging structures, and operational procedures that allow different V2G components, such as EVs, chargers, and grid systems, to interact efficiently and effectively. By reducing integration costs and simplifying deployment, standardization fosters flexibility and scalability, helping to ensure that V2G systems should be easily adapted to operate successfully in varying grid conditions, user needs, and regional regulations. Ultimately, interoperability supports broader market participation and allows V2G systems to function cohesively within diverse energy landscapes, contributing to enhanced grid stability, efficiency, and sustainability.

4.2.2.3 Real-time data exchange and control

V2G systems require real-time data exchange and advanced control capabilities to effectively respond to the dynamic and often unpredictable nature of grid conditions and energy market demands. Communication protocols in V2G systems shall support high-speed, low-latency data transfer to help ensure that adjustments in charging and discharging rates should be rapidly made. This capability is critical to enable V2G systems to provide key grid services such as frequency regulation, voltage support, and demand response. Real-time data exchange allows for continuous monitoring of important parameters such as EV battery levels, grid stability, grid load, and user preferences. These systems should optimize energy flow, helping to ensure that the grid remains balanced while at the same time maintaining EV availability for the owner's use. Furthermore,

advanced control mechanisms within these protocols enable automated decision-making processes based on real-time grid needs and energy market signals, such as price fluctuations or emergency conditions. This makes the system more responsive and adaptable, allowing for the continuous and automatic adjustment of charging or discharging actions in alignment with market or grid requirements. These capabilities enhance the system's flexibility, making V2G systems not only more effective in contributing to grid stability but also in enabling a more resilient and efficient energy ecosystem. The combination of real-time monitoring, automated control, and rapid response helps ensure that V2G systems should provide meaningful contributions to grid operations while optimizing the balance between energy supply, demand, and EV user needs.

4.2.2.4 Scalability and flexibility

To accommodate the growing number of EVs and charging stations, V2G communication protocols shall be designed with consideration to high scalability, helping to ensure that they should seamlessly integrate across a wide range of deployment scales, from individual EVs and small local systems to large-scale, city-wide networks. As the number of EVs on roads increases, the complexity of managing their charging and discharging shall also increase, and this shall be done in a way that benefits both users and grid operators. Scalable protocols are essential to handle this growth, enabling V2G systems to efficiently manage varying levels of participation, from individual vehicles to large aggregations of fleets of EVs. This scalability allows V2G systems to provide meaningful contributions to grid stability, energy efficiency, and renewable energy integration, regardless of the size or scope of the deployment.

Moreover, the scalability of V2G communication protocols shall help ensure that they should effectively support distributed and hierarchical control structures. At the local level, these systems should allow for autonomous decision-making by individual EVs or smaller aggregators, enabling them to respond quickly to real-time grid fluctuations, such as voltage or frequency deviations, without requiring centralized control. Localized decision-making helps ensure that V2G systems should react to micro-level grid conditions in a timely and efficient manner, for example adjusting charging and discharging rates based on immediate grid needs or user preferences. As the scale of the V2G deployment increases, however, it becomes important to integrate these local decisions into a higher-level coordination framework. This hierarchical structure allows for centralized management and optimization of V2G operations across broader geographic regions, such as cities or entire power networks, where decisions should need to account for macro-level grid stability, large-scale demand response strategies, or system-wide energy storage optimization.

The ability to coordinate locally and globally within a hierarchical framework makes V2G systems highly adaptable and flexible, allowing them to scale effectively with the growing demand for electric mobility. By providing localized autonomy, V2G systems should quickly respond to short-term fluctuations in grid conditions, helping to ensure that the system should absorb renewable energy, support frequency regulation, and balance supply and demand. At the same time, higher-level coordination helps ensure that V2G systems operate in harmony with the broader energy ecosystem, optimizing grid stability and efficiency on a large scale. This scalability and adaptability are critical as grid complexity and energy requirements evolve. As the number of EVs and charging stations continue to rise, scalable communication protocols help ensure that V2G systems remain reliable, responsive, and efficient, even when they operate in increasingly complex and dynamic environments. Ultimately, scalable V2G protocols shall be key to unlocking the full potential of electric vehicles as flexible, reliable assets for grid services, while supporting the transition to smarter, more sustainable energy systems.

4.2.2.5 Secure data exchange and privacy protection

As V2G systems involve extensive data exchange between EVs, charging stations, grid operators, and other stakeholders, robust cybersecurity measures are essential to safeguard both user privacy and the integrity of grid operations. Communication protocols in V2G systems shall be designed with built-in encryption standards to help ensure that all data exchanged between devices is securely transmitted and protected from unauthorized access. Authentication mechanisms are equally important, helping to ensure that only verified entities—such as authorized users, grid operators, or service providers—are granted access to control or

modify V2G system operations. Access control policies should be implemented to restrict access based on predefined roles, helping to ensure that each participant in the V2G network should only access the data or functions that are necessary for their roles, thereby reducing potential security vulnerabilities.

In addition to safeguarding against external threats, secure data exchange is crucial to help prevent malicious actions, such as unauthorized manipulation of charging or discharging operations, which could disrupt grid stability or/and lead to financial losses. By enforcing strong encryption, authentication, and access control, these protocols reduce the risks associated with cyberattacks, such as data breaches, denial-of-service attacks, or the compromise of sensitive operational data.

Furthermore, privacy protection measures shall be integrated into V2G communication protocols to safeguard user information, such as charging patterns, location data, and financial transactions. User data should be anonymized wherever possible, helping to ensure that individuals cannot be identified from the data exchanged between devices. This approach mitigates concerns associated with data tracking and personal information disclosure, which could undermine user trust in V2G systems. Privacy protection is not only critical for maintaining user confidence but also for helping to ensure compliance with data protection regulations, such as the General Data Protection Regulation (GDPR) in Europe or similar local policies. By anonymizing and securing user data, V2G systems should foster a trust-based relationship with users, encouraging broader participation while adhering to regulatory standards.

Ultimately, robust cybersecurity and privacy measures are vital to the success and scalability of V2G systems. By helping to ensure secure, authorized access and protecting sensitive information, these measures enhance the reliability and trustworthiness of V2G technology, enabling its broader adoption and integration into modern smart grids.

4.2.2.6 Adaptive communication for grid services

V2G communication protocols should be designed to support diverse grid services, such as frequency regulation, peak load management, and energy storage. These services require protocols that should adapt to varying operational demands, including predictive energy flows, time-sensitive responses, and varying load conditions. For instance, during peak demand, protocols should facilitate the discharge of stored EV energy back to the grid, while during periods of high renewable energy generation, they should support charging from the grid. Adaptive communication enables V2G systems to operate effectively within dynamic grid environments, thereby offering a range of services tailored to current grid needs.

4.2.2.7 Efficient aggregation and coordination

V2G protocols should facilitate the aggregation of multiple EVs into coordinated units or VPPs, allowing smaller contributions to be pooled for greater grid impact. Efficient aggregation protocols enable aggregators to coordinate charging schedules, manage capacity, and participate in energy and ancillary service markets. By aggregating resources, V2G systems should provide grid services on a scale, making small EV contributions meaningful within the broader energy network. Protocols should support flexible scheduling, load forecasting, and market-based bidding to optimize aggregated resources for grid services, helping to ensure reliable participation in grid operations.

4.2.2.8 Compatibility with market-based pricing

To foster active participation in V2G systems, communication protocols shall be compatible with market-based pricing mechanisms, enabling EV owners to receive fair compensation for their contributions to grid support services. V2G protocols should facilitate the exchange of dynamic pricing signals, such as time-of-use (ToU) rates or real-time market prices, directly to EVs or aggregators (P. Chen et al. [B14]). This capability helps ensure that charging and discharging behaviors should be adjusted in real time, aligning them with prevailing market conditions and grid demands. For example, during peak demand periods, higher compensation rates might incentivize EV owners to discharge energy into the grid, while off-peak hours might offer lower rates for charging, allowing for more efficient use of available energy resources.

The integration of dynamic pricing within V2G systems offers significant flexibility, enabling EV owners to increase economic returns by adjusting their participation according to market conditions. This should create an attractive financial incentive for EV owners to engage in grid services, whether by providing frequency regulation, demand response, or voltage support. Moreover, by compensating EV owners based on the value they provide to the grid, V2G systems become financially viable for participants, enhancing the system's overall sustainability and encouraging broader adoption. As a result, the integration of market-based pricing into V2G communication protocols not only improves grid efficiency but also transforms EVs into valuable assets for both users and grid operators, driving the growth of V2G technology.

4.2.3 Advanced communication systems

4.2.3.1 General

Advanced communication systems are a critical component of V2G infrastructure, enabling high-speed, secure, and reliable data exchange among EVs, chargers, grid operators, and other stakeholders. As V2G systems scale and become more integrated with the power grid, communication systems need to evolve to support real-time control, dynamic energy management, and large-scale coordination. Below is an outline of the advanced communication requirements and technologies necessary to optimize V2G operations.

4.2.3.2 Real-time communication and low-latency data transfer

V2G systems require real-time communications to respond promptly to grid demands, enabling rapid adjustments in charging and discharging patterns. Low-latency data transfer is crucial for services like frequency regulation, demand response, and voltage support, where even slight delays could impact on grid stability. Technologies such as 5G and fiber-optic networks provide high bandwidth and low latency needed for real-time V2G applications, allowing EVs to send and receive data with minimal delay.

Additionally, edge computing should play a significant role by processing data closer to the source (such as EVs and chargers), reducing the reliance on centralized servers and enabling faster decision-making. By integrating edge computing with advanced communication networks, V2G systems should enhance responsiveness, reduce latency, and improve overall efficiency in grid interactions.

Additionally, the emergence of sixth-generation (6G) wireless communication technologies is poised to significantly enhance the performance and capabilities of V2G systems. Building upon the foundations of 5G, 6G is expected to offer ultra-low latency (< 0.1 ms), ultra-high reliability, massive device connectivity, and data rates exceeding 1 Tbps, which shall be crucial in supporting real-time energy trading, precise load control, and autonomous coordination among distributed EVs. In V2G frameworks that require hierarchical control or decentralized orchestration, 6G's intelligent edge computing and integrated sensing-communication-control mechanisms should facilitate seamless interaction between vehicles, aggregators, and grid entities. Furthermore, 6G's support for AI-native network functions enables adaptive learning from energy demand and supply data, thus improving forecasting accuracy and operational resilience. To fully realize these advantages, future V2G system standards shall consider compatibility with 6G architecture, including network slicing for service isolation, semantic-aware communication protocols, and ultra-dense heterogeneous network deployment across charging infrastructure.

4.2.3.3 Machine-to-machine (M2M) communication

Machine-to-machine (M2M) communication enables direct interaction among devices within the V2G network, such as EVs, charging stations, and control systems, without the need for human intervention. M2M communication is essential for autonomous V2G operations, allowing devices to automatically exchange data, adjust power flows, and respond to grid signals (Moloisane, Malekian, and Bogatinoska [B53]). M2M technology enhances operational efficiency, as it allows V2G systems to function with minimal human input, optimizing energy exchange and enabling fast, automated responses.

M2M communication is particularly useful in aggregated V2G setups, where multiple EVs operate as a VPP. Through M2M protocols, the VPP should efficiently manage and coordinate the energy contributions of multiple EVs, helping to ensure that aggregated resources align with grid requirements. This capability is essential for scaling V2G systems and enabling their participation in grid services at a larger scale.

4.2.3.4 IoT integration and device connectivity

The Internet of Things (IoT) is instrumental in connecting and managing the vast number of devices involved in V2G operations. IoT platforms enable seamless connectivity between EVs, chargers, aggregators, and grid operators, allowing for unified data collection, monitoring, and control. IoT integration supports real-time data tracking, providing detailed information on EV battery status, grid conditions, and user preferences, which helps optimize energy flows and maintain grid stability.

Advanced IoT platforms should also enhance the flexibility and scalability of V2G systems by supporting a wide variety of devices and communication protocols, facilitating interoperability across different manufacturers and regions. With IoT, V2G systems should become more adaptive, allowing for remote updates and diagnostics, predictive maintenance, and optimized energy management based on real-time insights.

4.2.3.5 Cybersecurity in communication systems

Given the sensitivity of data involved in V2G systems, robust cybersecurity measures are essential for advanced communication systems. Cybersecurity protocols shall protect the systems against unauthorized access, data breaches, and potential cyber-attacks that could compromise grid operations or user privacy. Encryption, firewalls, and secure authentication are standard security components, but more advanced measures including blockchain technology and zero-trust architecture, should further enhance data protection.

Blockchain, for instance, should enable secure, tamper-proof transaction records between EVs, aggregators, and grid operators, reducing the risk of fraud or unauthorized access. A zero-trust approach, which requires continuous verification of users and devices, helps ensure that only trusted entities should access V2G systems, thereby reducing the risk of cyber threats. As V2G systems scale up, these cybersecurity measures are crucial for maintaining user trust and helping to ensure the reliability of V2G operations.

4.2.3.6 Artificial intelligence (AI) and machine learning (ML) for predictive control

AI and ML technologies enable predictive control and smart decision-making in V2G communication systems. AI algorithms should analyze large data sets from EVs, chargers, and grid sensors to forecast grid demands, user behaviors, and renewable energy availability. This predictive capability allows V2G systems to optimize energy exchange based on future grid needs, increasing efficiency and reducing costs.

ML models should also dynamically adjust V2G operations to align with time-of-use pricing, user schedules, and renewable energy fluctuations. By learning from historical data, these systems should predict the best times to charge or discharge EVs, enhancing their participation in grid services like demand response. Predictive control also helps balance grid supply and demand, making V2G systems more adaptable and effective in complex, changing environments.

4.2.3.7 Cloud computing and data analytics

Cloud computing provides a centralized platform to manage the vast amounts of data generated by V2G systems, supporting data storage, analysis, and remote access (Tao, Ota, and Dong [B79]). Data analytics on cloud platforms allow V2G stakeholders to gain insights into grid conditions, EV usage patterns, and energy market trends, which should be used to improve system performance and inform decision-making.

Cloud-based systems also enable remote control of V2G infrastructure, allowing operators to monitor and manage V2G devices across large geographical areas. With the scalability of cloud platforms, V2G systems should handle increasing numbers of connected devices and users, supporting widespread adoption and larger-

scale operations. Combined with advanced analytics, cloud computing enables V2G systems to operate with greater efficiency, predictability, and reliability.

4.2.3.8 Advanced protocols for hierarchical and decentralized control

To support the hierarchical structure of V2G systems—ranging from individual EVs to TSOs—advanced communication protocols should enable both centralized and decentralized control options. In hierarchical control, protocols facilitate coordinated decision-making across different levels, such as device-level control for individual EVs, aggregator-level coordination, and grid-level oversight.

For decentralized V2G setups, protocols should support peer-to-peer (P2P) communication between devices, allowing for localized control and self-management of V2G resources. Decentralized communication systems enhance system resilience, as they reduce reliance on a single central authority, that results in consequential bottlenecks or/and single points of failure. The flexibility of these protocols supports V2G scalability, enabling the system to adapt to various grids and market conditions while maintaining optimal performance.

4.2.4 BMS

4.2.4.1 General

The BMS is a crucial component in V2G systems, as it oversees the safe and efficient operation of the EV battery, optimizing it for both mobility and grid support. The BMS is responsible for monitoring, controlling, and safeguarding the battery's health, lifespan, and performance—key factors when EVs are utilized as distributed energy resources in grid operations. Below is a detailed exploration of the BMS functionalities and requirements for effective V2G integration.

4.2.4.2 Real-time battery monitoring and diagnostics

A primary function of the BMS in V2G applications is real-time monitoring of the battery's status, which involves tracking key parameters such as voltage, current, temperature, and SoC. In the context of V2G, where the EV battery is actively used to discharge power back to the grid, continuous and precise monitoring is critical to help ensure safe operation and prevention of equipment damage. Overcharging, deep discharge, or thermal runaway are potential risks that might compromise both the battery's safety and its performance. The BMS plays a key role in detecting these detrimental events in real time and taking preventive actions, such as adjusting charging or discharging rates, issuing alerts, or initiating protective measures like disconnecting the battery from the grid to help prevent further damage.

The BMS is furthermore responsible for monitoring the State of Health (SoH) of the battery, which is crucial for understanding the battery's long-term performance and helping to ensure that it operates within safe limits. By tracking the battery's SoH, the BMS should balance the need to increase energy output for grid services with the necessity of preserving the battery's longevity. This balance is especially important in V2G systems, where batteries should undergo frequent charging and discharging cycles as they provide services such as frequency regulation or demand response.

Advanced diagnostic algorithms embedded within the BMS further enhance its functionality by enabling it to predict battery degradation over time. These algorithms use historical data, real-time performance metrics, and environmental factors to forecast when the battery should experience a reduction in capacity or efficiency. By predicting potential degradation, the BMS should adjust the charging and discharging cycles dynamically, helping to ensure that the battery remains in an optimal condition while still responding to grid demands. This approach helps optimize the battery's life cycle, reducing the frequency of costly replacements and helping to ensure that the V2G system remains cost-effective and reliable throughout its operational lifetime.

4.2.4.3 Charge and discharge control

To enable bidirectional energy flow in V2G systems, the BMS shall precisely manage both charging and discharging processes, dynamically adjusting the rate of energy exchange based on grid signals, market conditions, and user requirements. During peak grid demand, the BMS controls the discharge rate, allowing the battery to supply energy to the grid while helping to ensure that the battery's health and performance are not compromised. By carefully managing the discharge process, the BMS helps ensure that the battery operates within safe limits, helping to prevent deep discharge, overheating, or other factors that could negatively impact the battery's longevity. Conversely, during low demand periods or when there is an excess supply of renewable energy, the BMS regulates the charging process, storing energy in the battery for later use while optimizing the charging rate to maintain battery health and help ensure the battery is sufficiently charged for future grid support.

The BMS employs adaptive charging algorithms to optimize energy flow, adjusting the charging current to help avoid overheating and reduce cycle-related wear, which is particularly important in V2G systems where frequent charging and discharging cycles are normal. These algorithms help ensure that the charging process is efficient and controlled, considering the battery's temperature, charge level, and SoH. By managing the charging current in real-time, the BMS helps to balance energy demands with battery longevity, helping to prevent excessive wear and tear caused by frequent cycling.

Furthermore, the BMS incorporates smart charging algorithms that should prioritize charging during off-peak hours, aligning with time-of-use pricing models. By charging during periods of low electricity demand (often at night), the BMS reduces energy costs for EV owners and reduces the strain on the grid during peak periods. This functionality not only benefits EV owners by reducing their energy costs but also helps to optimize grid operations, as the V2G system should be managed to respond to grid needs while maintaining economic efficiency.

4.2.4.4 Thermal management

Thermal management is a critical component for maintaining the health and longevity of batteries, especially in V2G systems, where frequent charging and discharging cycles should generate significant heat (Gupta et al. [B28]). The BMS integrates temperature monitoring and control systems that continuously track the battery's temperature during charge and discharge operations. By closely monitoring temperature, the BMS helps ensure that the battery remains within safe operational limits. If the temperature rises beyond a predefined threshold, the BMS should activate cooling mechanisms, such as fans, liquid cooling, or thermal sinks, to dissipate excess heat and help prevent potential thermal damage that could degrade the battery's performance or compromise its safety.

In addition to protecting the battery from overheating, proper thermal management plays a vital role in optimizing the charging efficiency and mitigating the performance loss that should occur due to temperature fluctuations. As temperatures affect the battery's internal resistance, uncontrolled temperature increases or decreases should lead to inefficiencies and increased cycle wear. By maintaining the battery within an optimal temperature range, the BMS helps ensure that charge and discharge processes are more efficient, leading to faster charging times and reduced energy losses.

In colder environments, where low temperatures should hinder battery performance, the BMS should also incorporate pre-heating functions. Before initiating a charge or discharge, the BMS should activate mechanisms to warm up the battery to an optimal operating temperature, helping to ensure that the battery performs effectively. This helps ensure that low-temperature impacts, such as increased internal resistance and reduced available energy, are mitigated, allowing the V2G system to operate efficiently even in suboptimal environmental conditions.

4.2.4.5 State estimation algorithms

Accurate estimations of the battery's state are critical for V2G applications, as it helps ensure the battery is ready to deliver or receive energy, as needed. The BMS employs advanced algorithms to calculate various states, including SoC, SoH, and SoP, which together provide a comprehensive view of the battery's readiness for V2G operations (Y. Zhang et al. [B95]). SoC indicates the current energy level of the battery, SoH reflects the battery's overall condition, and SoP measures the available power capacity at a given time.

These state estimations are based on complex algorithms that incorporate historical and real-time data, making them more accurate and reliable over time. By providing precise information on the battery's readiness and capacity, these algorithms enable the BMS to manage V2G operations more effectively, helping to ensure that energy delivery is reliable and the battery is not overused or degraded unnecessarily.

4.2.4.6 Depth of discharge (DoD) control

The DoD is a critical parameter in managing battery wear and extending battery lifespan, especially in V2G operations where batteries frequently cycle between charging and discharging. Higher DoD levels, where a significant portion of the battery's capacity is utilized during each discharge cycle, should accelerate battery degradation by increasing stress on the battery cells. To mitigate this, the BMS should set limits on DoD, controlling the amount of energy that is withdrawn from the battery during each cycle. By restricting the DoD, the BMS helps ensure that only a portion of the battery's total capacity is used, helping to preserve the health of the battery and reduce the rate of capacity loss over time.

The BMS should also optimize the DoD in response to varying grid needs and the battery's SoH, for example, during periods of high grid demand, the BMS should allow a deeper discharge to contribute more energy to the grid, thereby supporting grid stability and meeting market requirements. Conversely, during normal operations or when the battery's health is at risk, the BMS might limit the DoD to reduce wear and help ensure the battery is not over-stressed. By adjusting the DoD based on both real-time grid requirements and battery conditions, the BMS should balance the need for immediate grid support with the goal of long-term battery preservation.

This flexibility is key to helping to ensure that the battery should effectively support V2G services without compromising its longevity. By actively managing the DoD, the BMS optimizes the battery's service life, reducing the frequency of costly replacements and maintaining the system's reliability and performance over time. In this way, the BMS plays an integral role in sustaining the economic viability and reliability of V2G systems, helping to ensure that EVs should provide grid services efficiently while reducing the impact on battery degradation.

4.2.5 Compatible hardware and infrastructure

4.2.5.1 General

The successful deployment of V2G systems requires compatible hardware and infrastructure capable of supporting bidirectional energy flow and securing reliable communication with the grid. This infrastructure includes bidirectional chargers, EV batteries, communication devices, grid integration equipment, and strategically placed charging stations. Together, these elements enable V2G systems to function as distributed energy resources, providing grid services like frequency regulation, peak load reduction, and renewable energy support. Below is a detailed overview of the key hardware and infrastructure components required to help ensure V2G compatibility.

4.2.5.2 Bidirectional chargers

Bidirectional chargers are critical components of V2G systems, enabling the flow of energy in both directions—allowing EVs to not only draw power from the grid but also supply energy back into the grid. These chargers should be either ac or dc, but dc chargers are typically preferred in V2G applications due to

their ability to control power flows with enhanced precision and efficiency. dc chargers enable higher charging speeds, reduced energy conversion losses, and better overall efficiency, which is crucial for increasing the effectiveness of energy transactions in V2G operations. Given their direct conversion from dc to dc (EV battery to grid), dc chargers reduce the need for multiple stages of energy conversion, resulting in fewer energy losses and faster response times.

Key requirements for bidirectional chargers in V2G systems include high power conversion efficiency, reliable grid synchronization, and real-time data exchange capabilities. Chargers shall efficiently manage power flow while maintaining synchronization with the grid, helping to ensure that energy flow is stable and that the V2G system contributes effectively to grid services, such as frequency regulation, voltage support, or demand response. Real-time data exchange is essential for coordinating the charging and discharging processes, helping to ensure that the EVs' battery levels, grid conditions, and market signals are continuously monitored and optimized for both grid stability and user preferences.

Safety standards are another critical consideration for bidirectional chargers, as they shall operate safely under grid-connected conditions. These chargers shall incorporate mechanisms to help prevent overload, short circuits, or any fault conditions that could disrupt the grid or damage the EV's battery system. Compliance with established safety standards, such as IEC 61851 for EV charging systems and UL 1741 for grid-tied inverters, is essential to help ensure safe operation across different environments and grid conditions.

Additionally, bidirectional chargers shall include communication interfaces that support standardized interoperability protocols such as described in Mültin [B56] or Garofalaki et al. [B27]. These interfaces enable seamless integration with V2G-compatible EVs, charging stations, and central management systems. Standardized communication helps ensure that chargers should work with a variety of EVs, regardless of manufacturer or model, and allows for centralized control, monitoring, and optimization of V2G operations. By meeting these requirements, bidirectional chargers play a pivotal role in helping to ensure that V2G systems should operate efficiently, safely, and in coordination with grid operators and other stakeholders.

4.2.5.3 Communication and data management devices

Effective communication devices are essential for the successful operation of V2G systems, enabling seamless, real-time data exchange between EVs, charging stations, and grid operators. These communication devices shall support low-latency, high-speed data transfer to help ensure that the system should respond immediately to grid signals, such as requests for frequency regulation, voltage support, or demand response. Timely communication between all V2G components is crucial for maintaining grid stability, as delayed responses might lead to inefficiency or even grid instability. Communication devices shall also be fully compatible with V2G communication protocols and standards—such as in Mültin [B56] and Garofalaki et al. [B27], and other interoperability frameworks—helping to ensure that data is exchanged efficiently across diverse systems and equipment provided by various manufacturers. This compatibility fosters a cohesive V2G ecosystem, making it easier to integrate new technologies, devices, and participants into existing grid operations.

In addition to robust communication devices, data management infrastructure plays a pivotal role in enabling the efficient and reliable operation of V2G systems. The infrastructure shall be capable of storing, processing, and analyzing large volumes of data generated by EVs, such as battery levels, charging patterns, and availability, as well as real-time grid conditions, including voltage, frequency, and demand. Advanced cloud-based platforms and edge computing solutions are particularly well-suited for V2G systems, as they enable distributed data management and processing closer to the point of use. Edge computing allows for the real-time analysis of data on-site, reducing the time required for decision-making and enabling faster, more accurate responses to grid fluctuations or user preferences.

Cloud-based solutions further support scalability by providing a centralized framework for aggregating and analyzing data from large numbers of V2G devices, enhancing system-wide optimization and coordination. Together, these solutions improve response times and reduce the dependency on centralized control systems, thus enhancing the scalability and efficiency of V2G operations. As the number of connected EVs and charging

stations grows, these decentralized data management solutions help ensure that V2G systems remain agile, reliable, and capable of scaling to meet the demands of larger deployments. By incorporating both advanced communication devices and data management infrastructure, V2G systems should achieve a high level of operational flexibility, performance, and integration with grid services.

4.2.5.4 Grid integration equipment

Integrating V2G systems with the grid requires specialized equipment capable of managing bidirectional power flows while helping to ensure grid stability. Inverters and grid-tied converters are critical components in this integration, as they convert the dc power stored in EV batteries into ac power that should be fed back into the grid. These devices shall be highly efficient to reduce power losses during the conversion process, as any inefficiency could reduce the effectiveness of V2G systems in providing essential grid services. High conversion efficiency is crucial not only for helping to ensure that energy is delivered to the grid with minimal loss but also for optimizing the overall performance of the V2G system, including its ability to support frequency regulation, voltage stabilization, and demand response.

Additionally, grid integration equipment shall support grid synchronization, which helps ensure that V2G systems maintain appropriate alignment with the power grid's voltage, frequency, and phase. This synchronization is vital to help prevent disruptions and maintain stable grid operations, as any misalignment could cause imbalances, damage, or even power outages. The synchronization process requires precise control systems of the inverters and converters, helping to ensure that energy delivered from EVs does not interfere with the existing grid dynamics but instead complements the grid's needs.

In some V2G applications, particularly those involving high power levels or large-scale deployments, additional equipment such as transformers and power conditioners should be required to accommodate the increased power flows. Transformers adjust the voltage levels for safe and efficient transmission of power, while power conditioners help maintain the quality of the electricity by filtering out any disturbances or harmonic distortions that could affect the grid or connected loads. Furthermore, protective relays and circuit breakers play an essential role in helping to ensure the safety of both the grid and the V2G system. These devices help prevent overloading and fault conditions by quickly disconnecting the V2G system from the grid in the event of anomalies, such as short circuits, voltage surges, or other fault scenarios. By providing immediate protection, they safeguard the system from potential damage and help ensure the overall reliability of grid operations.

Thus, the successful integration of V2G systems requires a combination of highly efficient inverters, grid synchronization mechanisms, and robust safety features, along with additional equipment to handle high power levels and maintain grid quality. This infrastructure helps ensure that V2G systems should contribute effectively to grid stability while operating safely and efficiently within the larger energy ecosystem.

4.2.5.5 Charging station infrastructure

The strategic placement of charging stations is essential for increasing the benefits of V2G systems. Stations should be in areas with high EV traffic, such as urban centers, workplaces, public parking facilities, and residential complexes, to help ensure that EVs have convenient access to charging and discharging facilities. This infrastructure not only supports V2G participation but also encourages EV adoption by providing reliable, easily accessible charging options.

Public and workplace charging stations play an important role in supporting aggregation efforts, since multiple EVs should be charged and discharged in coordinated schedules to meet grid demands. These stations should be equipped with bidirectional charging capabilities, reliable communication interfaces, and real-time monitoring systems to facilitate efficient grid support. Residential chargers with V2G functionality also provide benefits by allowing homeowners to participate “in demand” responses and other grid services, supporting local grid stability.

4.2.6 Energy management software

4.2.6.1 General

The use of energy management software (EMS) is essential for the efficient and coordinated operation of V2G systems, thus enabling real-time monitoring, data analytics, control, and decision-making. EMS facilitates the interaction between EVs, grid operators, and aggregators, optimizing the charging and discharging of EV batteries to support grid stability, increasing economic returns, and helping to ensure user convenience. This software serves as the central brain of V2G systems, balancing the grid's needs with user demands and battery health. An overview of the key functions and requirements for EMS in V2G applications is presented below.

4.2.6.2 Real-time monitoring and control

Real-time monitoring and control are core functions of EMS, allowing it to gather data on EV battery levels, grid conditions, and energy market prices. EMS continuously tracks parameters such as SoC, SoH, and grid demand, by adjusting charging and discharging rates in response to grid signals. This real-time monitoring helps ensure V2G resources should respond quickly to grid needs, providing services like frequency regulation, peak load reduction, and voltage support.

The control functions within EMS allow it to execute complex energy strategies autonomously, aligning V2G operations with the grid's dynamic requirements. For instance, during peak demand, EMS should automatically initiate discharge from EV batteries, supplying stored energy to the grid while preserving battery health by following predefined operational thresholds. Real-time control also enables EMS to halt or delay charging when grid capacity is strained, thereby reducing stress on the system and maintaining grid stability.

4.2.6.3 Predictive analytics and load forecasting

Predictive analytics in EMS leverages ML and AI algorithms to forecast future energy demand, user charging patterns, and renewable energy availability. These predictions enable EMS to proactively schedule charging and discharging cycles, helping to ensure V2G resources are available when the grid needs them most. For instance, EMS should predict peak demand hours based on historical data and adjust charging schedules accordingly to help avoid costly peak tariffs and enhance grid support.

Load forecasting also allows EMS to balance energy storage with user mobility needs, helping to ensure EVs are charged sufficiently for users' planned journeys while still providing backup power for the grid. By forecasting renewable energy availability, such as solar or wind power, EMS should prioritize charging during periods of excess renewable generation, enhancing the sustainability of V2G operations.

4.2.6.4 Optimization of charging and discharging schedules

EMS optimizes charging and discharging schedules to balance grid support, economic benefits, and battery longevity. By analyzing ToU pricing and dynamic pricing signals, EMS determines the best times to charge, or discharge based on energy costs and market opportunities. During low-cost periods, EMS should prioritize charging, storing energy for later discharge during high-cost, high-demand times. This schedule optimization helps reduce the energy expenses of EV owners and supports grid stability by strategically distributing energy when demand is high.

Battery health and degradation are also key considerations, as frequent cycling should impact battery lifespan. EMS uses battery management strategies to limit deep discharges and control the rate of charge, increasing battery life while still enabling active V2G participation. This optimization helps ensure that V2G operations remain economically viable and that batteries are preserved therefore supporting long-term use.

4.2.6.5 Aggregation and VPP management

Aggregation capabilities within EMS allow multiple EVs to operate collectively as a VPP, pooling their stored energy to provide grid services at scale. By coordinating the charging and discharging schedules of individual EVs, EMS increases the aggregated energy resource's impact, making it a valuable grid asset. This VPP management function enables even small, distributed V2G resources to participate in energy markets and grid services, such as frequency regulation and reserve power options.

The EMS for VPP management should generate market bids based on aggregated capacity, allowing V2G resources to participate in wholesale, capacity, and ancillary service markets. The software also tracks the performance of individual EVs within the VPP, helping to ensure each vehicle's contributions are accounted for and that grid support remains reliable and efficient.

4.2.6.6 Integration with market-based pricing and incentive programs

EMS facilitates V2G's participation in various energy markets by enabling market-based pricing and incentive program integration. By receiving real-time pricing signals and incentive data, EMS should make informed decisions on charging and discharging to increase revenue. This functionality allows EV owners to earn income from their V2G contributions, whether through dynamic pricing, time-of-use tariffs, or participation in capacity and ancillary markets.

EMS also provides user access to incentive programs, such as rebates for off-peak charging or compensation for providing grid services. By participating in these programs, EV owners should offset operational costs, such as battery wear, thereby making V2G an attractive option. Market integration within EMS helps ensure that V2G resources remain flexible and responsive, adapting their operations based on both grid needs and economic signals.

4.2.6.7 User interface and customization

An intuitive user interface (UI) within EMS enables EV owners to monitor and control their V2G participation, providing transparency and flexibility. Through the UI, users should view battery levels, upcoming charging schedules, earnings from grid participation, and environmental impacts of their V2G activities. This visibility empowers EV owners to make informed choices and feel more engaged in supporting the grid.

Customization options allow users to set preferences for V2G participation, such as limiting discharge rates, setting minimum charge levels, or choosing preferred charging times. These settings help ensure that V2G operations align with user convenience and mobility needs while still providing valuable grid support. User customization fosters greater adoption of the process by allowing flexibility in V2G engagement.

4.3 System network interaction and communication

4.3.1 Intelligent charging and discharging

4.3.1.1 General

It is suggested that V2G systems should implement intelligent charging and discharging functionalities to optimize energy flows based on real-time grid conditions, user-defined parameters, and economic incentives. These functions need to be supported by predictive analytics, adaptive control mechanisms, and demand-response strategies to help ensure effective grid participation while maintaining vehicle availability and battery health.

4.3.1.2 Adaptive charging based on grid signals

Charging and discharging operations in V2G systems need to be regulated in response to grid conditions, prioritizing energy efficiency and grid reliability. During periods of low grid demand or high renewable energy generation, charging rates should be increased to absorb excess electricity, helping to prevent energy waste and supporting renewable integration. Conversely, during peak demand periods, charging should be reduced or paused, and stored energy should be discharged to alleviate grid power congestion. These operations shall be coordinated with real-time grid signals, including time-of-use pricing, frequency regulation requests, and renewable generation forecasts, to help ensure that V2G-enabled vehicles contribute effectively to grid balancing.

4.3.1.3 Predictive and adaptive scheduling

To enhance energy optimization, V2G systems employ predictive analytics to anticipate grid demand fluctuations, renewable energy output variations, and user charging behavior. Machine learning algorithms and historical data analysis should be used to determine the optimal timing for charging and discharging, helping to ensure that EVs store energy when renewable generation is at its peak and discharge energy when grid demand is highest. This proactive approach allows V2G systems to increase efficiency while reducing dependency on fossil-fuel-based power generation.

4.3.1.4 Demand-response driven discharging

V2G-enabled EVs should participate in demand-response programs that adjust their charging and discharging schedules in response to grid conditions. During peak load periods, discharging needs to be activated to supply energy to the grid, thereby reducing stress on generation assets and mitigating the risk of power shortages. V2G control systems should prioritize load balancing by coordinating discharging operations across multiple vehicles, helping to ensure that no single area of the distribution network is disproportionately affected. Aggregators should manage these discharges strategically, scheduling energy injections in anticipation of peak demand rather than reacting solely to real-time grid signals.

4.3.1.5 User-centric control and customization

The implementation of intelligent charging and discharging shall consider user mobility requirements and battery longevity. EV owners should have the ability to configure charging preferences, including minimum state-of-charge thresholds and depth-of-discharge limits, to help ensure that their vehicles remain available for transportation needs. Battery management algorithms should regulate charge rates, temperature, and cycling frequency to reduce degradation while maintaining active participation in V2G services. These measures shall extend battery life and help ensure that V2G-enabled vehicles remain a reliable and sustainable energy resource.

4.3.1.6 Battery health and lifespan management

To manage the impact of frequent cycling on battery health, intelligent charging and discharging systems include battery management algorithms that reduce degradation. By controlling the DoD, adjusting charge rates, and helping to prevent overcharging, these systems preserve battery health while still allowing active V2G participation.

The software optimizes DoD limits based on current grid conditions and user-defined thresholds, helping to ensure discharging only occurs under safe working conditions. This capability not only extends the battery life but also enhances the reliability of V2G resources, making EVs a more sustainable and economically viable grid asset.

4.3.1.7 Vehicle-to-home (V2H) and vehicle-to-building (V2B) capabilities

Beyond supporting the grid, intelligent charging and discharging systems should provide energy to local loads, such as homes or buildings, through vehicle-to-home (V2H) or vehicle-to-building (V2B) capabilities. During outages or periods of high electricity prices, the system should discharge power from the EV to support critical appliances or/and reduce building electricity costs.

These capabilities enhance energy resilience of individual users and facilities, allowing them to rely on their EVs as backup power. By supporting both local and grid-level demands, intelligent charging and discharging systems make vehicle-to-grid (V2G) systems versatile, contributing to both macro- and micro-level energy management.

4.3.1.8 Compliance and implementation

V2G systems should adhere to established grid interconnection and communication standards, including ISO/DIS 15118-2:2018-06 and IEEE Std 2030.5-2023, to help ensure interoperability between EVs, charging infrastructure, and grid operators. Control algorithms shall be regularly updated to integrate advancements in forecasting models, adaptive pricing mechanisms, and enhanced battery management strategies. Aggregators and system operators shall coordinate V2G participation to maintain grid stability while helping to ensure that the financial and operational viability of EV owners is preserved.

By implementing these requirements, V2G systems shall enhance grid reliability, optimize renewable energy utilization, and support sustainable energy management while helping to ensure that EV participation remains economically and operationally feasible.

4.3.2 Vehicle-to-grid energy trading

4.3.2.1 Market structure

V2G energy trading operates within a multi-tiered electricity market structure where EVs function as DERs, participating in wholesale energy markets, capacity markets, and ancillary service markets. The market structure defines the roles of grid operators, aggregators, energy suppliers, and EV owners, helping to ensure efficient coordination of energy flows, pricing mechanisms, and regulatory compliance.

The V2G market is divided into the following segments:

- *Wholesale electricity market:* EVs participate in large-scale electricity trading, responding to real-time price fluctuations and usually selling stored energy when market prices are high.
- *Capacity market:* EVs commit to providing power during peak demand events, supporting grid reliability by serving as dispatchable energy reserves.
- *Ancillary service market:* EVs provide frequency regulation, voltage support, and spinning reserves, responding to grid fluctuations to maintain stability.
- *Retail and P2P energy trading:* EV owners trade energy within local microgrids, communities, or through blockchain-enabled platforms, bypassing traditional utilities.

Market participation is facilitated by aggregators, which consolidate multiple EVs into VPPs, optimizing their energy contributions while helping to ensure compliance with grid requirements. The integration of real-time market pricing, demand-response mechanisms, and supervisory control systems enables seamless V2G transactions.

4.3.2.2 Pricing mechanisms

4.3.2.2.1 General

V2G energy trading should operate within a dynamic pricing framework that aligns economic incentives, grid stability, and market efficiency. Pricing mechanisms should optimize charging and discharging schedules, enabling EV owners to increase revenue while helping to ensure grid flexibility and efficient renewable energy utilization. These pricing structures should integrate real-time data analytics, predictive algorithms, and market-based decision-making to align energy trading with supply-demand conditions.

4.3.2.2.2 ToU Pricing

ToU pricing should be implemented as a predefined electricity rate structure that varies based on the time of day, reflecting expected demand patterns and grid conditions. ToU pricing should be static and pre-scheduled, meaning rates are determined in advance by utilities, regulators, or market operators and do not fluctuate based on real-time market conditions. Pricing schedules should be publicly available, helping to ensure that EV owners should plan charging and discharging schedules efficiently.

4.3.2.2.3 Real-time market pricing (RTP)

Real-time market pricing (RTP) should be implemented to adjust electricity rates dynamically based on instantaneous supply and demand conditions in the wholesale electricity market. This pricing method should enable EVs to respond to real-time price fluctuations, optimizing charging and discharging strategies accordingly. RTP signals should be broadcast in real time through grid communication networks, enabling automated decision-making for charging and discharging operations. Market participation strategies should be implemented to help ensure grid reliability, with constraints applied to help prevent excessive discharges that could compromise system stability.

4.3.2.2.4 Day-ahead and intraday market pricing

To enhance trading efficiency and grid responsiveness, V2G systems should be capable of participating in day-ahead and intraday electricity markets as follows:

- *Day-ahead pricing:* EVs should submit bids for charging and discharging one day in advance based on forecasted electricity prices, demand levels, and renewable energy generation. Aggregators should optimize these bids to increase revenue while helping to ensure grid compliance.
- *Intraday pricing:* EVs should be capable of adjusting their charging and discharging schedules within the same trading day based on updated market conditions. Intraday pricing should provide flexibility for responding to unforeseen grid fluctuations and optimizing V2G energy transactions.

4.3.2.2.5 Demand response and incentive-based pricing

V2G systems should be integrated with demand response programs, enabling price-responsive energy dispatch that aligns with grid balancing needs. Pricing models for demand response participation should include the following:

- *Incentive-based pricing:* EV owners should receive direct financial incentives for participating in grid demand reduction events by adjusting their charging and discharging patterns as requested by the grid operator or aggregator.
- *Critical peak pricing (CPP):* During extreme peak demand periods, V2G-enabled EVs should be incentivized to discharge stored energy at higher compensation rates, reducing the need for conventional peaking power plants.

- *Automated load shifting:* Charging schedules should be automatically adjusted based on price signals to reduce grid congestion while increasing cost savings for EV owners.

4.3.2.2.6 P2P energy trading pricing

V2G systems should support P2P energy trading to facilitate direct transactions between EV owners and consumers within localized microgrids or decentralized energy networks. P2P pricing models should include the following:

- *Auction-based pricing:* EV owners should place bids to sell stored energy, and buyers (such as households or businesses) should place offers based on their energy needs. A market-clearing algorithm should match buyers and sellers dynamically, helping to ensure that transactions occur at an equilibrium price. Auctions should be conducted in real-time or scheduled intervals, based on local energy market structures.
- *Negotiation-based pricing:* EV owners and buyers should be allowed to set flexible pricing terms based on mutual agreements. Smart contract frameworks should automate negotiation and transaction execution to help ensure pricing transparency and reduce transaction delays. EV owners should have the option to prioritize selling energy based on price preferences, contract duration, or grid constraints.

4.3.2.3 Aggregation and VPP models

Individual EVs often have limited capacity, which should restrict their direct participation in large energy markets. Through aggregation and VPP models, multiple EVs should pool their stored energy to act as a single, significant power resource. Aggregators or VPP operators manage these pooled resources, coordinating charging and discharging schedules to optimize participation in energy markets.

By aggregating V2G resources, VPPs should meet the capacity requirements of wholesale, capacity, and ancillary service markets, enabling small contributions to provide grid services at scale. This aggregated model makes V2G energy trading viable on a larger scale, allowing multiple EVs to collectively support the grid, earn revenue, and participate in market opportunities that would otherwise be inaccessible for individual vehicles.

4.3.2.4 Smart contracts and blockchain for secure transactions

V2G energy trading involves numerous transactions, such as buying and selling energy and participating in ancillary services. Blockchain technology and smart contracts should facilitate these transactions securely and transparently, helping to ensure that all parties have a verifiable record of energy trades. Smart contracts automate the execution of energy trades based on predefined conditions, such as market prices or grid demand, reducing the need for intermediaries and helping to ensure timely, accurate payment of accounts.

Blockchain's decentralized nature enhances the transparency and security of V2G energy trading, helping to prevent fraud and helping to ensure that EV owners receive fair compensation for their contributions. By simplifying and securing transaction processes, blockchain technology supports scalable, efficient V2G trading systems, allowing for smooth, real-time interactions among EVs, aggregators, and grid operators.

4.3.2.5 User incentives and revenue generation

To encourage EV owners to participate in V2G energy trading, incentive mechanisms are essential. These should include dynamic pricing, tax rebates, or direct monetary compensation for providing energy to the grid. Additionally, specific incentives, such as lower electricity tariffs during off-peak charging, encourage users to adopt V2G practices that support grid needs.

By participating in V2G energy trading, EV owners should generate revenue through price arbitrage, frequency regulation, demand response, and other grid services. These financial benefits help offset the operational costs of V2G, such as battery wear, making V2G an economically viable choice. Over time, as the V2G market grows, increased competition and incentive structures are likely to further enhance these financial opportunities, making V2G energy trading attractive to a larger community.

4.3.2.6 Regulatory compliance and market integration

V2G energy trading operates within a regulated environment, and compliance with market rules and interconnection standards is crucial. Regulations define eligibility for market participation, setting requirements for response time, capacity, and data security. By adhering to these standards, V2G systems should participate reliably in energy markets, helping to ensure that energy contributions align with grid needs and regulatory requirements.

Market integration is also necessary to facilitate V2G trading, helping to ensure compatibility with existing energy market platforms and policies. By complying with market regulations, V2G systems should enable smoother interactions with grid operators, enhancing system reliability and fostering trust among stakeholders.

To help ensure system stability, market fairness, and cybersecurity, V2G energy trading shall operate within a structured regulatory framework. Key supervisory requirements include the following:

- Grid interconnection standards (e.g., ISO/DIS 15118-2:2018–06, IEEE Std 2030.5-2023) to facilitate compatibility with existing infrastructure.
- Market oversight mechanisms to help prevent energy imbalances and market manipulation.
- Cybersecurity measures to protect transaction data and help prevent unauthorized access.

Through real-time monitoring platforms, grid operators maintain oversight of aggregated V2G resources, helping to ensure compliance with grid stability and market regulations.

4.3.3 Condition monitoring and control of vehicles and energy systems

4.3.3.1 General

Condition monitoring and control are essential for the reliable and efficient operation of V2G systems, as they help ensure that both EVs and energy infrastructure maintain optimal performance while meeting grid demands. By tracking the health, usage, and environmental conditions of EV batteries, chargers, and grid connections, condition monitoring systems enable proactive maintenance and adaptive control, increasing V2G system efficiency and longevity. Below is an overview of the key components, technologies, and benefits of condition monitoring and control in V2G applications.

4.3.3.2 Real-time monitoring of battery health and performance

The BMS within each EV continuously monitors battery health by tracking critical parameters such as SoC, SoH, temperature, and voltage. Real-time monitoring allows the system to detect signs of degradation or abnormal operation, helping to ensure that the battery remains within safe operating conditions. For V2G applications, this real-time data is vital for scheduling charging and discharging based on battery capacity, health, and grid needs.

Condition monitoring of battery health helps prevent overuse or overheating, which should lead to reduced lifespan or failure. By maintaining battery integrity, the system helps ensure that EVs are available for grid support and meet user mobility requirements. Data from battery health monitoring should also support predictive maintenance, extending battery life by identifying issues before they detrimentally impact on performance.

4.3.3.3 Charger and grid connection monitoring

The chargers and grid connections used in V2G systems require constant monitoring to help ensure reliable and safe operations. This includes tracking power flow rates, temperature, and equipment performance. Chargers should operate in various conditions and at different power levels, making it essential that they are monitored for signs of wear, overheating, or malfunction. By identifying potential issues early, the system should perform maintenance or/and adjustments to help avoid interruptions to V2G operations.

Monitoring grid connections is also critical, as bidirectional power flow should introduce new challenges, such as voltage instability or detrimental power quality issues. Real-time monitoring helps identify these issues before they escalate, allowing V2G systems to adjust power flows and protect grid infrastructure. Condition monitoring helps ensure that V2G chargers and grid connections operate efficiently, supporting both local and system-wide grid stability.

4.3.3.4 Predictive maintenance for battery and equipment longevity

Predictive maintenance uses data analytics and machine learning to forecast potential equipment failures and/or degradation, allowing V2G operators to take preventive actions before problems occur. By analyzing historical data and real-time monitoring insights, predictive algorithms should identify patterns that signal wear, overheating, or other characteristics associated with poor battery health or charger performance.

Predictive maintenance helps reduce unexpected downtimes and costly repairs, helping to ensure that EVs remain available for V2G participation and that chargers and grid connections operate reliably. For example, if the system detects that a battery's charge retention is declining, it should adjust charging schedules to reduce strain or recommend effective maintenance actions. This approach extends equipment life and reduces operational costs, making V2G systems more sustainable and economically viable.

4.3.3.5 Adaptive control for optimized energy flow

Adaptive control systems within V2G platforms allow dynamic adjustment of charging and discharging rates, adapting to real-time grid conditions, market prices, and battery health. For example, during peak demand, the system should prioritize discharging energy to the grid at controlled rates to help avoid battery strain while still supporting grid needs. Adaptive control also enables EVs to charge more aggressively when renewable energy availability is high, thereby increasing the use of clean energy and reducing grid dependency on fossil fuels.

Adaptive control functions enable V2G systems to respond quickly to grid demands and market conditions without compromising battery integrity. By adjusting energy flows based on real-time conditions, these systems optimize V2G participation, making the technology more flexible and responsive to changing grid dynamics.

4.3.3.6 Environmental and operational condition monitoring

Environmental factors, such as temperature and humidity, should have a detrimental impact on the performance and lifespan of batteries, chargers, and other V2G equipment. Condition monitoring systems track these environmental conditions to help prevent overheating, freezing, or other adverse impacts. For example, if high ambient temperatures are detected, the system shall usually activate cooling mechanisms or reduce the charging rate to help avoid thermal damage.

Operational conditions, such as unusual usage patterns and load cycles, are also monitored to inform adaptive control strategies and predict maintenance needs. Monitoring these factors helps V2G systems maintain consistent performance and help ensure that environmental or operational factors do not negatively impact equipment reliability or efficiency.

4.3.3.7 Data analytics for condition reporting and decision support

Data analytics in condition monitoring systems enables report generation on the performance, health, and usage of V2G components. These reports should provide insights into energy contributions, battery degradation, and equipment wear over time, supporting decision-making for maintenance, upgrades, or operational adjustments. Condition reports allow EV owners, aggregators, and grid operators to understand the performance of V2G resources and allow them to make informed decisions that optimize both grid support and asset longevity.

These insights also support strategic planning, as historical data on battery usage, charger performance, and grid interactions should guide long-term improvements to V2G infrastructure and policy development. Decision support derived from data analytics helps ensure that V2G systems evolve in response to real-world usage patterns and performance data.

4.3.3.8 Communication and integration with EMS

Condition monitoring systems are integrated with EMS, which coordinates real-time data and controls energy flow based on grid and market conditions. This integration enables the seamless execution of intelligent charging and discharging strategies, adaptive control, and predictive maintenance. By sharing data between condition monitoring systems and EMS, V2G operations should be optimized to align with grid requirements, user preferences, and equipment health.

For instance, if condition monitoring detects that an EV battery is reaching a predefined discharge limit, it should communicate with the EMS to halt discharging and initiate charging based on available grid capacity or renewable generation. This communication helps ensure that V2G operations remain responsive, reliable, and sustainable.

4.3.3.9 Safety protocols and fault detection

Safety is a priority in V2G systems, and condition monitoring includes protocols for fault detection and safety management. The system constantly checks for anomalies, such as abnormal temperature increases, voltage fluctuations, or power surges, and initiates safety protocols when necessary. These safety measures include shutting down equipment, isolating faulty components, and alerting operators to potential hazards.

Fault detection helps prevent potential issues from escalating into failures that could impact grid stability or user safety. By implementing strict safety protocols and continuous monitoring, V2G systems reduce risks associated with high-power charging and discharging, thereby protecting both infrastructure and users.

4.3.4 Communication network

4.3.4.1 General

The communication network is a vital component of V2G systems, facilitating real-time data exchange among EVs, charging stations, grid operators, and aggregators. This network enables seamless coordination of charging and discharging operations, demand response, grid balancing, and participation in energy markets. A robust communication network in V2G helps ensure efficient, secure, and reliable interactions that increase both grid support and user convenience. A detailed overview of the key aspects and requirements for V2G communication networks follows now. Recent advancements in wireless and digital communication technologies, particularly the development of 6G networks, are expected to play a transformative role in V2G systems. As EV adoption increases and V2G networks scale in complexity, next-generation communication infrastructure shall support massive connectivity, ultra-reliable low-latency communication (URLLC), and real-time distributed intelligence. These features are particularly relevant to V2G environments where dynamic coordination among large numbers of EVs, chargers, and control centers is required. Through integrated sensing and communication capabilities, 6G enables real-time visibility into EV battery status, grid conditions, and user preferences, enhancing situational awareness and enabling predictive control.

4.3.4.2 Hierarchical and decentralized communication structures

V2G systems should use either hierarchical or decentralized communication structures, depending on operational requirements and scale. In a hierarchical model, communication flows through several levels, such as device-level control for EVs, aggregator-level coordination, and grid-level management. This structure is beneficial for centralized decision-making and offers a structured approach to large-scale V2G coordination.

In decentralized V2G setups, P2P communication is enabled, allowing devices to communicate directly and coordinate local energy exchange without relying on central authority. Decentralized communication structures reduce reliance on a single point of control, increasing system resilience and scalability, especially in DER environments.

4.3.4.3 Real-time data exchange and low-latency communication

To manage dynamic grid demands and support fast response capabilities, V2G systems rely on real-time data exchange with minimal latency. Low-latency communication allows EVs to respond to grid signals almost instantaneously, adjusting charging and discharging rates in line with grid needs. This quick response is critical for services such as frequency regulation, voltage support, and demand response, where even small delays should impact grid stability.

High-speed communication technologies, such as 5G and fiber-optic networks, are well-suited for V2G applications, providing the bandwidth and low latency required for real-time interaction. These technologies enable high-frequency data updates, helping to ensure that V2G systems operate smoothly and efficiently in dynamic grid conditions.

4.3.4.4 IoT Integration for enhanced connectivity

The IoT provides the connectivity infrastructure needed to link a vast number of V2G devices, including EVs, chargers, aggregators, and grid management systems. IoT integration enables continuous data flow and monitoring, allowing V2G systems to operate as a unified network that responds dynamically to grid needs.

IoT platforms support remote monitoring and control, predictive maintenance, and real-time data analytics. By leveraging IoT, V2G communication networks achieve greater scalability and flexibility, making it possible to manage large fleets of EVs and a complex network of chargers and grid connections efficiently.

4.3.4.5 Edge computing for decentralized data processing

Edge computing enhances V2G communication by processing data closer to the source, such as within EVs or charging stations. By distributing data processing across multiple locations, edge computing reduces the load on central servers and reduces latency, allowing for faster decision-making and response times in V2G operations.

In a decentralized V2G setup, edge computing allows real-time processing of data on battery status, grid conditions, and user preferences, enabling EVs to adjust charging or discharging patterns instantaneously. This capability is particularly useful in high-density V2G networks, where reducing latency is critical for maintaining grid stability and efficiency.

4.3.4.6 Cybersecurity and data privacy

The V2G communication network involves extensive data exchange, including sensitive user information, energy transactions, and grid control data, making cybersecurity and data privacy essential. V2G networks incorporate robust security protocols, such as encryption, authentication, and access control, to help prevent unauthorized access and protect data integrity.

Blockchain technology should enhance security by providing transparent, tamper-proof records of V2G transactions, helping to ensure that only authorized entities have access. Additionally, privacy-preserving techniques should anonymize user data, helping to prevent sensitive information from being exposed. These security measures help foster user trust and help ensure that V2G systems contribute to grid stability without compromising data security.

4.3.4.7 Integration with EMS

The V2G communication network is integrated with EMS, which coordinates data flow and control functions. This integration allows EMS to receive real-time updates on battery status, grid conditions, and pricing signals, enabling adaptive control of charging and discharging based on both user needs and grid demands.

Through EMS, V2G systems should manage energy flows more effectively, optimize participation in energy markets, and provide grid services such as frequency regulation. The EMS integration streamlines communication across all V2G components, helping to ensure that each system functions cohesively as part of the larger grid network.

4.3.5 Communication modes and algorithms

4.3.5.1 General

In V2G systems, efficient communication modes and algorithms are essential for enabling EVs, charging stations, grid operators, and aggregators to interact dynamically and effectively. Communication modes determine how data flows through the V2G network, while algorithms facilitate adaptive control, optimization, and decision-making based on real-time conditions. Together, they help ensure V2G systems operate reliably, meeting both grid and user needs while increasing economic and environmental benefits. Below is an overview of key communication modes and algorithms in V2G systems.

4.3.5.2 Centralized communication mode

In a centralized communication mode, a central control unit or coordination center manages all V2G activities. The central system collects real-time data from EVs, charging stations, aggregators, and grid operators, and uses this information to make system-wide decisions regarding energy flow, demand response, and grid services. This mode is particularly useful for applications requiring a unified view of the V2G network, such as large-scale demand response, peak load reduction, and frequency regulation.

Centralized communication provides robust oversight and enables high-level optimization. However, it should introduce bottlenecks and scalability challenges, especially as the number of participating EVs grows. Centralized control algorithms, such as optimal power flow (OPF) and model predictive control (MPC), are often used to optimize system-wide energy flows and help ensure stability across the V2G network.

4.3.5.3 Distributed communication mode

In a distributed communication mode, individual entities within the V2G networks such as EVs, charging stations, and aggregators, make decisions based on local data and conditions. Distributed communication reduces reliance on a central authority, making the system more scalable and resilient to single points of failure. Each entity communicates with nearby devices, often forming a peer-to-peer network, to coordinate charging and discharging based on local grid needs and user requirements.

Distributed algorithms, such as consensus algorithms and multi-agent reinforcement learning (MARL), are used in this mode to enable cooperative decision-making across the network. These algorithms allow EVs and chargers to achieve grid objectives, like load balancing or voltage support, while making independent decisions based on local information. This mode is highly adaptable to complex grid environments, allowing V2G systems to operate efficiently even in decentralized grids.

4.3.5.4 Hybrid communication mode

A hybrid communication mode combines centralized and distributed elements, allowing for flexible and responsive V2G operations. In this mode, high-level decisions (e.g., overall energy dispatch and market participation) are made by a central authority, while local decisions (e.g., individual EV charging schedules and battery health management) are made at the device or aggregator level. Hybrid communication leverages the strengths of both centralized and distributed modes, optimizing system-wide objectives while allowing local flexibility.

Hierarchical control algorithms, such as two-level optimization and decomposition-based algorithms, are commonly used in hybrid modes to coordinate interactions between central and local levels. These algorithms enable high-level optimization without sacrificing the adaptability and resilience offered by distributed control.

4.3.5.5 P2P communication mode

In P2P communication, individual EVs and chargers communicate directly with each other, coordinating energy exchange and demand responses without the need for a central coordinator. This decentralized structure is ideal for applications where local energy sharing or microgrid operation is required. P2P communication supports flexibility and reduces the load on central servers, making it highly suitable for DER environments.

P2P algorithms, such as blockchain-based smart contracts and game-theoretic algorithms, are commonly used in this mode. These algorithms facilitate secure, automated transactions and fair energy exchange among V2G participants. Blockchain helps ensure data integrity and transparency, allowing participants to trade energy autonomously while maintaining a trusted record of transactions.

4.4 Typical scenarios

4.4.1 Microgrid

In microgrid scenarios, V2G technology plays a critical role in enhancing resilience, supporting renewable energy integration, and optimizing local energy management. A microgrid is a localized energy system that should operate autonomously (in “islanded mode”) or connect to the main grid. When operating in island mode, V2G allows EVs to act as mobile energy storage units, providing backup power to the microgrid during outages or in isolated areas. In these scenarios, V2G-enabled EVs discharge stored energy to maintain essential services—such as lighting, refrigeration, and communication systems—until grid power is restored. This capability is especially valuable in remote or underserved regions where traditional backup generation options are limited, helping the microgrid sustain reliable operations even during unexpected disruptions.

In microgrids with significant renewable energy integration, from sources such as solar and wind, V2G further supports energy stability by absorbing excess generation during peak production periods and discharging it when renewable output is low. For instance, during sunny or windy days when solar panels or wind turbines produce excess energy, V2G-enabled EVs should charge using this surplus, thereby helping to prevent energy waste and reducing the need for curtailment. This stored energy should later be discharged during times of low renewable generation, such as at night or during calm weather, maintaining a steady energy supply without relying on carbon-intensive sources. V2G thus enhances sustainability and efficiency of microgrids, helping them meet net-zero or low-emission goals.

A V2G pilot project in the UK represents a pioneering effort to integrate V2G technology into microgrid systems, with a focus on enhancing grid stability, optimizing the use of renewable energy, and improving the resilience of localized energy systems Trahand and Vare [B80]. The core objective of this project was to demonstrate how V2G-enabled EVs could support both renewable energy integration and emergency resilience, benefiting both the grid and EV owners. The project used bidirectional chargers that allow EVs to charge from the grid and return stored energy back into the grid when needed. This integration makes EVs

an essential part of the local energy system, particularly in microgrids. During periods of excess renewable generation, such as when solar panels produce more electricity than needed, the V2G-enabled EVs absorb this surplus energy, helping to prevent energy waste and reducing the need for curtailment. Alternatively, during times of low renewable generation or high demand, such as at night or during poor visibility weather, the EVs discharge their stored energy into the grid or microgrid, helping to stabilize the energy supply.

A key feature of the V2G pilot project is the ability of V2G-enabled EVs to act as mobile energy storage units during grid failures. In the event of an outage, the vehicles should provide electricity to critical loads such as lighting, refrigeration, and communication systems, helping to ensure the continuity of service even in remote or off-grid areas. This feature is particularly beneficial in geographical regions where traditional backup generation options are limited or unreliable. Furthermore, the V2G system helps manage peak loads by discharging stored energy during high consumption periods, such as hot summer afternoons or cold winter evenings, when demand spikes due to cooling or heating. By providing energy to the grid during these times, V2G helps reduce the strain on the grid and reduces the need for fossil-fuel-powered peaking plants, contributing to a cleaner, more sustainable grid.

In addition to the technical benefits identified, the project offers economic incentives for both EV owners and grid operators. EV owners are compensated for the energy they provide to the grid, making participation in V2G programs financially attractive. Meanwhile, grid operators benefit from reduced operational costs, as the V2G system helps avoid costly investments in additional infrastructure, such as backup generators or grid upgrades. V2G also increases energy efficiency, as it enables grid operators to better balance supply and demand without relying on expensive, carbon-intensive solutions. Ultimately, the V2G Pilot Project demonstrates the potential of V2G technology to support renewable energy integration, grid stability, and emergency resilience in microgrids, while offering economic and environmental benefits. It serves as a scalable model that should be adapted for future V2G-enabled microgrid projects worldwide, helping to pave the way for a more sustainable, resilient energy future.

4.4.2 Peak charging load regulation

In V2G systems, peak charging demand regulation is essential for managing high demand charging periods, helping to prevent excessive strain on the grid, and helping to ensure efficient power distribution. As EV adoption grows, uncoordinated charging during peak hours should significantly increase grid load, leading to grid congestion, voltage instability, and the need for costly infrastructure upgrades. Without effective regulation, utilities should have to rely on expensive peaking power plants, which are inefficient and carbon intensive. By leveraging intelligent charging management and demand response mechanisms, V2G systems should optimize charging schedules, reducing grid stress and energy costs.

Peak charging demand typically occurs when many EVs charge simultaneously, particularly during late afternoon and evening hours, when residential electricity demand is already high. This problem is exacerbated in commercial fleet depots and urban areas, where multiple vehicles require rapid charging within limited time frames. Uncontrolled charging should overload distribution transformers, increase reliance on fossil-fuel-powered peaking plants, and cause voltage fluctuations, particularly in grids with high levels of renewable energy penetration. To maintain grid stability, it is essential that dynamic load balancing strategies are implemented that distribute charging demand more efficiently across different time periods.

To mitigate peak charging congestion, V2G-enabled systems incorporate smart charging technologies that dynamically adjust charging rates based on real-time grid conditions. Load management systems optimize charging by prioritizing off-peak charging sessions and spreading demand across available grid capacity. Through demand response programs, utilities should incentivize EV owners to shift their charging to lower-demand periods, thereby helping to prevent localized grid congestion. Additionally, aggregators play a key role in fleet coordination, helping to ensure that EVs connected to charging infrastructure operate in a way that aligns with grid constraints.

An advanced approach to peak charging regulation involves bidirectional energy flow through V2G technology, where EVs should discharge stored energy back to the grid during high-demand periods. This capability enables EVs to act as decentralized energy resources, effectively offsetting their own charging impact and stabilizing voltage levels in areas with high EV penetration. Energy management platforms utilize machine learning algorithms to forecast grid conditions and optimize EV charging and discharging schedules accordingly. By integrating these control mechanisms, V2G systems enhance grid reliability and reduce dependence on additional generation capacity.

A project conducted by a certain consultancy in 2019 assessed the feasibility and economic viability of V2G technology in the UK (V2GB [B81]). The study found that residential V2G charging could become economically viable in the near term. A practical demonstration of V2G's potential was conducted in the UK, where a fleet of EVs equipped with bidirectional charging capabilities provided grid services by discharging stored energy during peak demand periods. This real-world use case illustrated how V2G technology could enhance grid stability, reduce reliance on traditional peaking power plants, and offer economic benefits to EV owners through participation in energy markets. Since solar power is unavailable overnight and flexible generation is reduced by 35% in high-flexibility scenarios, there is insufficient energy to fully charge EVs overnight. As a result, V2G utilization during the day is constrained, limiting its ability to discharge stored energy to the grid. This highlights a key challenge in charging peak regulation, where the effectiveness of V2G depends on the availability of renewable generation and grid flexibility. Without adequate overnight charging, V2G cannot fully smooth out daytime peak demand, underscoring the need for coordinated charging strategies that align with renewable generation patterns and grid conditions.

4.4.3 Energy supplier standby capacity

In V2G systems, EVs should serve as an on-demand standby capacity for energy suppliers, providing a flexible and responsive backup that enhances grid stability. Standby capacity refers to the reserve energy resources that should be quickly called upon during times of unexpected demand surges, generation shortfalls, or emergencies. Traditionally, this capacity is provided by standby generators or peaker plants, which are often costly and less environmentally friendly. However, with V2G, EVs collectively act as a mobile, distributed energy reserve that energy suppliers should rely on to stabilize the grid in real time.

When V2G-enabled EVs are connected to the grid, they should offer standby capacity by being on standby to discharge stored energy when the grid requires immediate support. This capability is particularly valuable in scenarios involving unexpected demand spikes or shortfalls in renewable generation, such as when solar or wind output drops suddenly due to unfavorable weather changes. V2G allows energy suppliers to tap into the stored energy in EV batteries to meet this shortfall without activating more carbon-intensive resources, thereby improving grid efficiency and reducing emissions.

From an economic perspective, V2G-enabled standby capacity also benefits EV owners, who should receive compensation for making their vehicles available as a reserve. This financial incentive offsets battery wear from cycling and encourages more participation in V2G programs. Moreover, standby capacity from V2G provides flexibility that traditional power plants cannot match, as V2G resources should be deployed quickly and adjusted dynamically according to real-time grid needs. In grids with high renewable penetration, this responsive standby capacity from V2G becomes even more crucial, as it supports the balance of intermittent renewable energy sources, reducing the need for fossil-fuel backups and moving toward a more resilient, low-carbon grid.

A related project serves as a strong case study for V2G as an energy supplier standby capacity, showcasing how EVs with bidirectional charging should support grid stability and flexibility in a net zero energy system ("Interflex Project Summary" [B37]). The study conducted by Energy Systems Catapult highlights that V2G could provide over 50 GWh of flexible capacity by 2050, reducing the need for grid-connected energy storage and peak-demand generation. Compared to unmanaged charging, V2G should significantly displace traditional electricity storage investments, while offering distributed energy reserves that should be activated in response to demand fluctuations. Additionally, V2G supports renewable energy integration by balancing

intermittency, helping to ensure higher utilization of renewable generation, and reducing reliance on high-cost flexible generation technologies. This demonstrates the potential for EVs to act as decentralized backup power sources, reinforcing grid resilience while optimizing electricity system operation.

4.4.4 Participating in ancillary services

In frequency regulation services, V2G-enabled EVs provide real-time adjustments to balance supply and demand, helping to ensure that grid frequency remains within safe operational limits. When generation exceeds demand, the grid frequency rises, and EVs should increase their charging rate to absorb excess energy, helping to prevent over-frequency conditions. Conversely, when demand exceeds generation, the frequency drops, and V2G systems should discharge stored energy to compensate for the shortfall.

V2G systems also contribute to voltage stability by providing or absorbing reactive power. Voltage fluctuations are common in distribution networks with high penetration of renewable energy, as intermittent sources like solar PV and wind turbines cause voltage variations. V2G-enabled EVs should help stabilize local voltage levels by dynamically adjusting their reactive power injections based on grid conditions. For instance, in urban areas with high EV adoption, multiple fast chargers operating simultaneously should create localized voltage drops. By integrating V2G-enabled reactive power compensation, EVs should inject reactive power during low-voltage conditions, supporting power factor correction and improving grid efficiency.

In contingency scenarios where unexpected power plant outages or renewable energy fluctuations occur, spinning reserves are required to restore grid balance. Traditionally, gas turbines and hydroelectric plants provide spinning reserves, but V2G technology offers a faster and cleaner alternative by deploying aggregated EV fleets to provide instant-response reserves.

A project conducted by an institute in Denmark from 2016 to 2019, demonstrated the capability of contemporary EVs to provide advanced smart grid services through V2G technology (P. B. Andersen et al. [B5]). In collaboration with other worldwide partners, the project utilized a fleet of EVs and V2G dc chargers to perform a series of tests and demonstrations. These activities took place both in a power laboratory facility and in a real-world field study, where EVs successfully provided frequency containment reserve (FCR) services to the Danish power grid. The project concluded that the tested EVs and charging infrastructure were technically capable of delivering all frequency regulation services used in Denmark, marking a significant step toward the commercialization of V2G technology.

5. V2G-based hierarchical partitioned aggregation

5.1 V2G system aggregation

5.1.1 General

Vehicle-to-grid (V2G) system aggregation entails the coordinated management of multiple electric vehicles (EVs) as a unified energy resource. This strategy enhances grid stability, optimizes energy utilization, and facilitates renewable energy integration by leveraging distributed EV resources (Yilmaz and Krein [B92]). Furthermore, the use cases about V2G-based hierarchical partitioned aggregation model could be referenced in [Annex A](#).

5.1.2 Aggregation methods

5.1.2.1 General

The aggregation method defines the processes and frameworks for managing EV clusters at different levels as described in the following:

5.1.2.2 Scope of aggregation

Specify the level of aggregation according to operational objectives as follows:

- *Local level:* Aggregate EVs within a confined area, such as residential communities or corporate campuses, to support local energy management.
- *Regional level:* Coordinate EVs across broader geographical areas to provide grid services, including load balancing and frequency regulation.
- *Virtual power plants (VPPs):* Integrate EVs with distributed energy resources to form VPPs capable of delivering ancillary services such as demand response and peak shaving.

5.1.2.3 Constraints

Address the constraints that impact aggregation effectiveness as follows:

- *Communication infrastructure:* Establish secure and reliable communication channels between EVs, aggregators, and grid operators (N. Matta et al. [B49]).
- *Regulatory compliance:* Adhere to national and regional standards governing V2G systems, including data privacy and security protocols.
- *Technical limitations:* Consider the variability in EV battery capacities, charging/discharging rates, and user availability.
- *Market participation:* Helping to ensure compatibility with market mechanisms and regulations for participation in energy markets (Y. Jin et al. [B40]).

5.1.3 Impacts of aggregation

Aggregated EVs deliver significant benefits to the power grid. Leverage the following advantages:

- a) *Grid stability:* Enable ancillary services, including frequency regulation and voltage support, to enhance grid reliability.
- b) *Peak load management:* Reduce peak demand by discharging EV batteries during high consumption periods and the need for additional generation capacity.
- c) *Renewable integration:* Absorb excess renewable energy during low-demand periods to mitigate curtailment.
- d) *Economic efficiency:* Facilitate energy market participation, generating revenue for EV owners and reducing system-wide energy costs.
- e) *Environmental benefits:* Improve sustainability by supporting efficient energy use and integrating renewable resources, reducing greenhouse gas emissions.

5.1.4 How to optimally aggregate

Achieving optimal aggregation requires a systematic approach. Implement the following practices:

- a) *Advanced algorithms:* Develop and deploy machine learning and optimization algorithms to predict EV availability, energy demand, and grid requirements, helping to ensure efficient scheduling (Le Floch, di Meglio, and S. Moura [B44]).
- b) *Hierarchical control structures:* Establish multi-tiered control systems to manage EV clusters at local, regional, and system-wide levels for scalability and responsiveness (M. Zhou et al. [B97]).

- c) *Robust communication protocols*: Use standardized, secure communication protocols for real-time data exchange between EVs, aggregators, and grid operators.
- d) *Incentive mechanisms*: Design pricing strategies and incentives to encourage EV owner participation in V2G programs.
- e) *Regulatory frameworks*: Collaborate with regulatory bodies to develop policies that support V2G aggregation and facilitate energy market access.
- f) *Battery management strategies*: Implement measures to reduce battery degradation, such as optimized charging cycles and state-of-health monitoring, to help ensure reliability and extend battery life.

5.2 Hierarchical aggregation techniques for EV

5.2.1 General

The integration of EVs into the power grid necessitates sophisticated aggregation techniques to manage their collective impact effectively. Hierarchical aggregation provides a structured approach for coordinating numerous EVs, enhancing grid stability, and optimizing energy utilization. Guidelines to help ensure efficient integration and operation are now described:

5.2.2 Aggregation algorithms and strategies

Effective aggregation of EVs requires advanced algorithms and strategies to manage their charging and discharging behaviors systematically. Expand and refine the following approaches:

- a) Hierarchical optimization:
 - 1) Implement multi-level optimization frameworks to balance individual EV user preferences with grid-wide operational requirements.
 - 2) Employ a two-level hierarchical optimization model. At the first level, increase user satisfaction by scheduling charging and discharging based on individual preferences. At the second level, help ensure grid constraints, such as power balance and voltage stability, are met (N. Bañol Arias et al. [B8]).
 - 3) Use distributed optimization techniques to enhance scalability and help ensure real-time responsiveness in large-scale EV deployments.
- b) Stochastic modeling:
 - 1) Incorporate stochastic models to address uncertainties in EV availability and energy demand effectively.
 - 2) Develop probabilistic models that capture variability in EV user behavior, such as departure and arrival times, charging needs, and usage patterns.
 - 3) Introduce a stochastic hierarchical framework that dynamically adjusts aggregation strategies based on real-time data, helping to ensure grid reliability during peak load scenarios and demand response events (T. Zhang et al. [B94]).
- c) Machine learning techniques:
 - 1) Leverage machine learning algorithms to predict EV charging patterns and optimize aggregation strategies dynamically.
 - 2) Train predictive models using historical data to forecast charging demands and identify high-usage periods accurately.
 - 3) Deploy reinforcement learning techniques to adaptively refine aggregation strategies in response to evolving grid conditions and user behaviors (S. Shahriar et al. [B77]).

By adopting these methods, the efficiency and reliability of EV aggregation systems should be enhanced. These frameworks need to be validated through simulations and pilot projects before widespread implementation. Robust communication protocols and data security measures to support the integration of these strategies into the grid infrastructure need to be established.

5.2.3 Data transmission and communication techniques

Reliable data transmission and communication systems are essential for the real-time coordination of aggregated electric vehicles (EVs). Advanced communication techniques to help ensure efficient and secure data exchange between EVs, aggregators, and the grid need to be implemented. The following strategies are recommended:

- a) Wireless access in vehicular environments (WAVE):
 - 1) Utilize the standard (IEEE Std 802.11p-2010 [B33]) for vehicle-to-everything (V2X) communication, enabling seamless data exchange between vehicles and infrastructure (Molina-Masegosa, Gozalvez, and Sepulcre [B52]).
 - 2) Establish low-latency communication channels to support real-time data transfer for functions such as load balancing and dynamic route planning.
 - 3) Optimize WAVE protocols to accommodate high-density EV environments, helping to ensure robust communication even in urban areas with heavy traffic.
 - 4) Integrate V2X systems with edge computing nodes to process data locally, reducing communication delays and enhancing scalability for large-scale EV networks.
- b) Power line communication (PLC):
 - 1) Employ existing electrical infrastructure for data transmission, leveraging the dual-use capability of power lines for communication.
 - 2) Develop PLC systems that support high-speed and noise-immune data exchange, helping to ensure reliability in environments with electromagnetic interference.
 - 3) Enhance PLC protocols to meet the specific demands of EV operations, such as monitoring charging status and controlling load distribution.
 - 4) Use PLC to facilitate bidirectional communication between EVs and grid operators, enabling dynamic updates of energy pricing and demand forecasts (C.-U. Park et al. [B62]).
- c) Simultaneous wireless power and data transfer (SWPDT):
 - 1) Integrate power and data transmission to streamline communication during EV charging processes.
 - 2) Design SWPDT systems with high-efficiency data carriers, helping to ensure stable and fast information exchange alongside power transfer (Y. Yao et al. [B91]).
 - 3) Implement plug-and-play toroidal-core inductors for injecting and extracting data carriers, reducing system complexity and setup requirements.
 - 4) Optimize SWPDT technology to help ensure compatibility with various wireless charging systems, enabling standardization across EV models and charging infrastructure (Z. Qian et al. [B65]).

5.2.4 Hierarchical network architecture

A hierarchical network architecture helps ensure efficient management of aggregated EVs by structuring communication and control across multiple levels are tailored to the specific objectives such as market participation, peak shaving, or other services. Different aggregation purposes necessitate varying scales of

EV fleets, and this architecture facilitates precise and timely data exchange for optimal coordination. The following hierarchical levels are expanded and refined to reflect these distinctions (see [Figure 1](#)):

- a) *Local level*—This level focuses on managing individual EVs for immediate, localized objectives such as peak shaving or small-scale grid support. It includes the following:
 - 1) *Communication protocols*: Develop low-latency, secure communication protocols between EVs and local infrastructure to help ensure seamless data exchange (Sultana, Basha, and Irfan [B78]).
 - 2) *Parameter monitoring*: Incorporate mechanisms to monitor EV-specific parameters like state of charge (SoC), location, and user charging preferences.
 - 3) *Intelligent charging systems*: Dynamically adjust charging rates based on real-time local grid conditions and user needs, enabling immediate response to localized demands.
 - 4) *Aggregation scope*: Typically manages small EV fleets sufficient for localized grid optimization, requiring less scalability compared to higher levels.
- b) *Regional level*—This level aggregates data from local levels to address broader objectives like regional peak shaving, load balancing, or limited market participation. It includes the following:
 - 1) *Data integration*: Design control algorithms that consolidate data from multiple local units, providing a regional energy demand overview and EV availability (X. Wu et al. [B87]).
 - 2) *Prediction models*: Utilize machine learning to forecast regional energy trends and optimize charging schedules, helping to ensure sufficient capacity during peak periods.
 - 3) *Load forecasting*: Integrate tools for accurate regional load forecasting to enhance operational resilience and address more significant fluctuations in demand.
 - 4) *Aggregation scope*: Manages moderate-sized EV fleets, enabling regional services like frequency regulation and demand response programs.

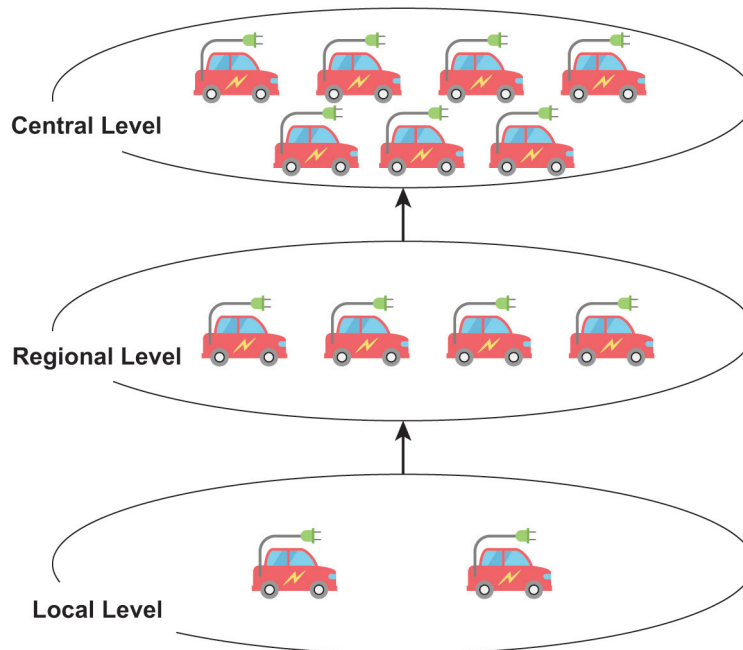


Figure 1—Hierarchical network architecture of EV aggregation

- c) *Central level*—This level coordinates the entire network for large-scale objectives such as wholesale market participation or inter-regional load balancing (J. Zhang et al. [B93]). It includes the following:
 - 1) *Centralized data management*: Implement a platform for real-time aggregation and processing of data from regional centers, helping to ensure a network-wide perspective.
 - 2) *Advanced optimization algorithms*: Optimize inter-regional energy flows using sophisticated algorithms to balance supply and demand across the network.
 - 3) *Critical oversight*: Enable centralized monitoring of metrics such as grid stability, fault detection, and system anomalies to help ensure rapid response during critical events.
 - 4) *Aggregation scope*: Requires extensive EV fleet coordination across regions, enabling services like ancillary market participation and large-scale grid stabilization (Lu and Wang [B47]).

5.2.5 Security and privacy protection

Helping to ensure security and privacy in EV aggregation is essential for protecting user data and maintaining system integrity. Comprehensive measures to address potential vulnerabilities and strengthen the reliability of EV aggregation systems need to be adopted. The following strategies to achieve robust protection need to be used and refined:

- a) Authentication protocols:
 - 1) Establish robust authentication mechanisms to verify the identity of devices and users reliably (W. Wang et al. [B85]).
 - 2) Implement multi-factor authentication protocols to enhance security, combining user credentials with device-specific factors such as cryptographic tokens or biometric identifiers.
 - 3) Develop protocols using digital signatures based on public key infrastructure (PKI) to help ensure secure and tamper-proof communication in consumer-centric EV charging environments.
 - 4) Regularly update authentication mechanisms to address emerging threats and help ensure compatibility with evolving cybersecurity standards.
- b) Data encryption:
 - 1) Utilize advanced encryption standards to safeguard data during transmission and storage (Jiang, Tao, and Guan [B39]).
 - 2) Adopt end-to-end encryption techniques to protect sensitive user and system data against interception or unauthorized access.
 - 3) Implement dynamic encryption key management systems to help ensure secure generation, distribution, and renewal of encryption keys.
 - 4) Use quantum-resistant encryption algorithms to future-proof communication against potential quantum computing threats.
- c) Access control:
 - 1) Define and enforce clear access policies to restrict data access to authorized entities only.
 - 2) Develop role-based access control (RBAC) systems that assign permissions based on predefined roles and responsibilities.
 - 3) Implement audit trails and logging mechanisms to monitor access attempts and identify unauthorized activities.
 - 4) Introduce adaptive access control mechanisms that dynamically adjust permissions based on contextual factors such as user behavior, location, and system status.

5.2.6 Best-in-class layering technologies

Adopting advanced layering technologies enhances the efficiency, reliability, and scalability of EV aggregation. These technologies support seamless communication, interoperability, and system-wide integration across heterogeneous infrastructure within the power grid. The following strategies are recommended to optimize EV management:

- a) Substation-oriented communication models:
 - 1) Implement communication layer models tailored for energy systems to support bidirectional interaction between EVs and grid operators. These models facilitate real-time control of charging and discharging processes and ensure interoperability across devices (Jiang, Tao, and Guan [B39]).
 - 2) Event-triggered messaging mechanisms should be used to enable low-latency responses during grid contingencies such as peak load or fault events (Aftab et al. [B2]).
 - 3) Applications should be extended to cover distributed energy resources (DERs) and EV assets, promoting flexible integration and automated coordination.
- b) Vehicular communication frameworks:
 - 1) Adopt wireless communication frameworks designed for mobility environments to enable secure, high-throughput, and low-latency data exchange. These frameworks should effectively support both vehicle-to-grid and vehicle-to-vehicle interactions in dynamic energy management scenarios (Jamborsalamati et al. [B38]).
 - 2) Cross-layer integration of vehicular and grid-side communication should form a unified framework that enhances operational reliability.
 - 3) Embedded security mechanisms, such as authenticated message exchange protocols, should be employed to safeguard data integrity and confidentiality during EV-grid interactions (Hussain et al. [B32]).
- c) Middleware-based integration solutions:
 - 1) Utilize middleware platforms (IEEE Std 1609.2™-2016 [B35]) as translation and coordination layers to bridge various communication protocols and system architectures.
 - 2) Design middleware frameworks capable of interfacing with both grid-oriented and vehicle-oriented communication models, ensuring compatibility across multiple vendors and platforms.
 - 3) Enable centralized data aggregation and distributed intelligence through these platforms to support predictive control, machine learning–based optimization, and real-time system feedback.
 - 4) Ensure middleware flexibility and upgradability to accommodate growing EV fleets and evolving infrastructure requirements.

5.3 Distributed EV resources aggregation

5.3.1 General

The aggregation of distributed electric vehicle (EV) resources is crucial for enhancing grid stability, optimizing energy utilization, and facilitating the integration of renewable energy sources. This section outlines methodologies and strategies for effective aggregation, focusing on regional resource assessment, flexible management, resource optimization, intelligent decision-making, and clustering models. The following approaches to help ensure robust aggregation practices need to be implemented.

5.3.2 Cluster model for electric vehicles

Implementing cluster models groups electric vehicles (EVs) based on specific characteristics, enhancing management efficiency and scalability in aggregation systems. These models enable tailored strategies for resource allocation, load balancing, and grid stability. Refine and adopt the following approaches to improve EV management:

- a) Geographical clustering:
 - 1) Group EVs based on their physical locations to manage local grid impacts more effectively.
 - 2) Divide geographical regions into manageable zones using grid topology and energy demand patterns as key factors.
 - 3) Develop algorithms to monitor and manage charging and discharging activities within each zone, reducing localized grid stress and helping to prevent congestion.
 - 4) Align geographical clusters with renewable energy generation zones, helping to ensure EV charging coincides with periods of high renewable availability to increase clean energy utilization.
- b) Behavioral clustering:
 - 1) Categorize EVs based on their usage patterns and charging behaviors to develop targeted management strategies.
 - 2) Use data analytics to classify EVs into behavioral groups, such as frequent commuters, occasional users, or heavy-duty vehicles.
 - 3) Design cluster-specific charging protocols that address unique behavioral needs, such as rapid charging for high-usage EVs and cost-efficient scheduling for occasional users.
 - 4) Implement predictive models within clusters to anticipate charging demand variations, enabling proactive resource allocation.
- c) Hierarchical clustering:
 - 1) Establish a multi-level clustering structure to facilitate scalability and control across large EV networks.
 - 2) Organize clusters hierarchically, where lower-level clusters (e.g., local groups) report to intermediate aggregators, which in turn connect to a central management system.
 - 3) Employ hierarchical distributed scheduling algorithms to balance loads within and across clusters, helping to ensure system-wide efficiency and reliability (X. Deng et al. [B18]).
 - 4) Validate hierarchical models using simulations or pilot studies, such as the proposed distributed EV charging scheduling method, which demonstrated scalability and control benefits.

5.3.3 Regional EV resource assessment and demand forecasting

Accurate assessment and forecasting of regional EV resources and their charging demands are essential for effective aggregation. Help ensure comprehensive planning and implementation through these steps:

- a) Data collection:
 - 1) Gather comprehensive and region-specific data on EV distribution, usage patterns, and charging behaviors (Hiry et al. [B30]).
 - 2) Establish automated data acquisition systems at charging stations to monitor real-time usage and energy consumption.

- 3) Integrate data sources such as traffic flow records, vehicle registrations, and user surveys to create a detailed EV operation profile.
- 4) Regularly update data sets to capture evolving trends in EV adoption and user behavior.
- b) Demand forecasting models:
 - 1) Employ advanced machine learning algorithms and statistical models to predict future charging demands with high accuracy.
 - 2) Train forecasting models using historical data, incorporating factors such as time of day, weather conditions, and local events that influence EV charging patterns (Rashid, Elfouly, and Chen [B67]).
 - 3) Use ensemble modeling techniques to combine multiple algorithms, enhancing prediction reliability.
 - 4) Integrate adaptive models capable of real-time updates to reflect changing conditions, such as sudden increases in EV adoption or infrastructure upgrades.
- c) Scenario analysis:
 - 1) Evaluate various scenarios to understand potential impacts on the grid under different conditions.
 - 2) Simulate the effects of varying EV adoption rates, including high-growth scenarios driven by policy incentives or market trends.
 - 3) Model the deployment of charging infrastructure, considering factors such as fast-charging stations, residential chargers, and public charging hubs.
 - 4) Assess the grid's capacity to handle extreme scenarios, such as simultaneous charging peaks or natural disasters affecting infrastructure (Botero and Rios [B10]).

5.3.4 Flexible charging and discharging management

Implementing flexible strategies for managing electric vehicle (EV) charging and discharging is essential for balancing grid load, optimizing resource efficiency, and supporting grid stability. The following approaches provide a robust framework for achieving these objectives.

- a) Smart charging protocols:
 - 1) Develop smart charging protocols that dynamically adjust charging rates based on real-time grid conditions and energy prices.
 - 2) Utilize advanced algorithms to monitor grid demand and renewable energy availability, helping to ensure charging rates align with current system capacity.
 - 3) Implement demand response mechanisms that incentivize users to charge during off-peak hours or periods of high renewable generation, reducing strain on the grid.
 - 4) Establish two-way communication systems between EVs and grid operators to enable real-time charging rate adjustments and efficient load management (Dahiwalé, Rather, and Mitra [B17]).
- b) Vehicle-to-grid (V2G) integration:
 - 1) Enable EVs to discharge energy back to the grid during peak demand periods, providing ancillary services and supporting grid reliability.
 - 2) Develop coordinated charging and discharging strategies through aggregators, allowing EV fleets to act as distributed energy resources.
 - 3) Facilitate frequency regulation, voltage support, and load balancing by leveraging V2G-enabled EVs.

- 4) Incorporate market-based mechanisms to encourage V2G participation, offering financial incentives for discharging during high-demand scenarios.
- c) User-centric approaches:
 - 1) Incorporate user preferences and mobility needs into charging schedules to help ensure convenience and participation.
 - 2) Design adaptive scheduling systems that consider user travel patterns, desired SoC, and departure times, reducing disruptions to daily activities.
 - 3) Provide mobile applications or interfaces that allow users to set preferences and receive notifications about optimal charging times and potential savings (Q. Wang et al. [B84]).
 - 4) Conduct user engagement campaigns to promote awareness of smart charging and V2G benefits, fostering higher adoption rates.

5.3.5 Resource allocation and optimization

Optimizing the allocation of aggregated electric vehicle (EV) resources is essential for enhancing grid performance, economic efficiency, and environmental sustainability. Implement strategies that help ensure effective resource utilization and contribute to a resilient energy infrastructure. Expand and refine the following approaches:

- a) *Optimization algorithms*—Utilize advanced optimization algorithms to determine the optimal charging and discharging schedules that reduce operational costs and greenhouse gas emissions:
 - 1) Develop algorithms capable of integrating multiple objectives, such as cost minimization, emission reduction, and grid stability, into a unified optimization framework (S. Mishra, A. Mondal, and S. Mondal [B51]).
 - 2) Implement a fast polytope-based approach to efficiently aggregate large-scale EV resources, reducing computational time while maintaining accuracy in schedule optimization.
 - 3) Incorporate real-time data from grid operators, weather forecasts, and market prices into optimization models to adapt schedules dynamically to changing conditions.
- b) *Load balancing techniques*—Distribute EV charging activities across different times and locations to help prevent grid congestion and maintain load stability:
 - 1) Employ predictive load balancing models that analyze historical and real-time data to forecast peak demand periods and distribute charging accordingly.
 - 2) Use geospatial analysis to identify regions with available grid capacity and redirect charging loads to help prevent local bottlenecks (X. Chen et al. [B15]).
 - 3) Coordinate load balancing efforts with renewable energy generation patterns, helping to ensure that EV charging aligns with periods of high renewable output.
- c) *Incentive mechanisms*—Design pricing strategies and incentives to encourage EV owners to participate in demand response programs and contribute to grid efficiency:
 - 1) Develop dynamic pricing models that offer reduced rates for charging during off-peak periods or times of excess renewable energy generation.
 - 2) Introduce financial incentives for EV owners who enable discharging during peak demand, supporting V2G initiatives and grid reliability.
 - 3) Create user-friendly platforms for incentive programs, helping to ensure transparency and accessibility to increase participation.

5.3.6 Collaborative learning and intelligent decision-making

Incorporating collaborative learning and intelligent decision-making processes enhances the adaptability, scalability, and efficiency of electric vehicle (EV) aggregation systems. These strategies enable dynamic resource management, improve forecasting accuracy, and support seamless communication among stakeholders. The following approaches outline key methodologies that are needed for implementation:

- a) *Distributed learning models*—Implement distributed learning models that enable EVs and aggregators to collaboratively learn from each other's data:
 - 1) Design federated learning frameworks to train machine learning models across distributed EV data sets without sharing raw data, preserving user privacy and data security (X. Zhou et al. [B98]).
 - 2) Use distributed models to improve demand forecasting accuracy by capturing regional and temporal variations in EV usage patterns.
 - 3) Integrate adaptive learning mechanisms that refine resource management strategies in response to changing grid conditions and user behaviors.
- b) *Artificial intelligence (AI) integration*—Apply AI techniques to analyze complex data sets and make real-time decisions regarding EV charging and discharging operations:
 - 1) Develop AI algorithms capable of processing high-dimensional data from diverse sources, such as grid demand, weather forecasts, and user preferences.
 - 2) Use reinforcement learning to optimize charging and discharging schedules, balancing grid stability with user convenience (Y. Cao et al. [B12]).
 - 3) Implement predictive analytics powered by AI to identify potential bottlenecks or inefficiencies in the aggregation system, enabling proactive solutions.
- c) *Collaborative platforms*—Develop collaborative platforms that facilitate seamless communication and data sharing among all stakeholders involved in EV aggregation:
 - 1) Create cloud-based platforms that integrate real-time data from EV owners, grid operators, and service providers, enabling coordinated decision-making.
 - 2) Design intuitive user interfaces that allow stakeholders to access relevant data, provide inputs, and monitor system performance.
 - 3) Use blockchain technology to help ensure data transparency, traceability, and security within the collaborative ecosystem.

6. Market-driven V2G aggregation and policy recommendations

6.1 V2G aggregation participation in energy markets

6.1.1 Framework

The framework of EV aggregator participating in energy market is shown in Figure 2. There are three components of the framework: the EV cluster, the EV aggregator, and the energy market. The energy market should be divided by time scale into real-time energy market, intraday energy market, day-ahead energy market, and medium- to long-term energy market. The EV cluster connected to the EV aggregator should be divided into two categories:

- a) *Dispatchable EVs*: The dispatchable EVs represent the EVs that accept the electricity price incentives determined by the EV aggregator. They are willing to participate in the energy market and be controlled by the EV aggregator to charge or discharge at the expense of consuming their battery life.

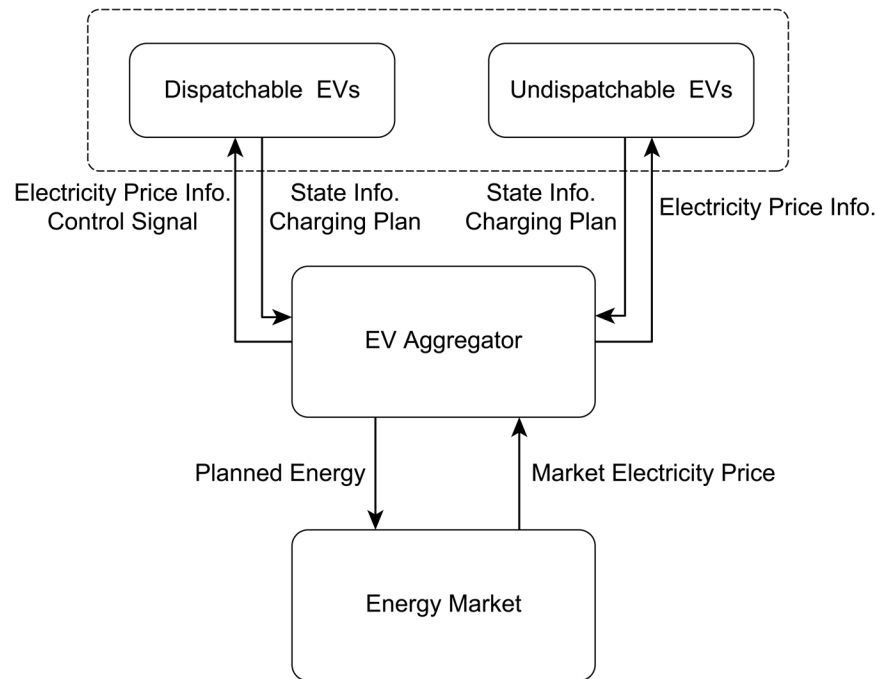


Figure 2—Framework of EV aggregator participating in energy market

- b) *Undispatchable EVs*: The undispatchable EVs represent the EVs that do not accept the electricity price incentives determined by the EV aggregator. They are unwilling to participate in the energy market at the current trading period, however, they should participate in the energy market at another trading period and be categorized to the dispatchable EVs if the electricity price incentives are accepted.

The interconnections between the EV cluster, EV aggregator, and energy market should be illustrated as follows:

- All the EVs should send their state information and charging plan to the EV aggregator first. The state information includes the SoC, capacity, maximum SoC, minimum SoC, and maximum charging/discharging power of their battery. The charging plan includes the starting time, ending time, and required SoC for charging their battery in the whole time. After receiving the electricity price information from the EV aggregator, they should decide whether to participate in the energy market. Then all the EVs should be categorized into two categories: dispatchable EVs and undispatchable EVs.
- The energy market should publish the market electricity price to the EV aggregator after receiving the planned energy from the EV aggregator before the trading period.
- The EV aggregator accepts the market electricity price published by the energy market to establish the electricity price incentives for all the EVs in the EV cluster. Furthermore, the EV aggregator sends the electricity price information, which includes the electricity price incentives determined by the EV aggregator and the market electricity price published by the energy market, to the EV cluster. The electricity price incentives should be used to guide the EVs to participate in the energy market. Then all the EVs in the EV cluster should decide whether to accept the electricity price incentives (To identify the dispatchable EVs and undispatchable EVs). Finally, the EV aggregator sends the control signal to the dispatchable EVs to control their battery charge or discharge according to the actual adjustment power in the energy market.

6.1.2 Incentive mechanism

6.1.2.1 Charging incentive mechanism

The EV aggregator should determine the charging electricity price according to the electricity price published by the energy market and the assessment of the planned energy of the EV cluster. All the EVs in the EV cluster should decide whether to participate in the energy market based on the charging electricity price determined by the EV aggregator and the market electricity price published by the energy market. If an EV accept the charging electricity price determined by the EV aggregator, it shall be categorized to the dispatchable EVs and should enjoy the charging prices and smart charging services published by the aggregator. The aggregator shall reasonably arrange the EV's charging process during the charging period, providing energy services to gain benefits by controlling the EV's charging. If an EV does not accept the charging electricity price determined by the EV aggregator, it means the EV is categorized to the undispachable EVs, and the EV is not under the aggregator's control, but instead connects to the grid for charging until the SoC reaches the user's charging plan. The charging electricity price should be calculated by Equation (1) as follows:

$$\rho_{cs}(t) = \alpha(t)\rho_{cm}(t) \quad (1)$$

where

ρ_{cs}	is the charging electricity price determined by the EV aggregator at time period t
$\rho_{cm}(t)$	is the charging electricity price published by the energy market at time period t
$\alpha(t)$	is a coefficient at time period t

6.1.2.2 Discharging incentive mechanism

The EV aggregator needs to set discharging electricity price to attract EV users to participate in discharging. In addition, to compensate for the loss of battery life caused by EV discharging, the EV aggregator shall also provide additional discharge compensation. The discharging electricity price should be calculated by Equation (2) as follows:

$$\rho_{ds}(t) = \rho_1(t) + \rho_0(t) \quad (2)$$

where

$\rho_{ds}(t)$	is the discharging electricity price determined by the EV aggregator at time period t
$\rho_1(t)$	is the discharge subsidy price at time period t
$\rho_0(t)$	is the compensation price for the loss of battery life at time period t

The compensation price $\rho_0(t)$ should be determined by the function of the depth of discharge and cycle count of an EV battery as follows:

$$\rho_0(t) = f(D, C) \quad (3)$$

where

D and C	are the depth of discharge and cycle count of an EV battery, respectively (L. Feichun et al. [B26])
$f(\cdot)$	is a mapping function

All the EVs in the EV cluster should decide to respond to one of the charging incentive mechanism, the discharging incentive mechanism, or both.

6.1.3 Objective functions of the EV aggregator

The EV aggregator regulates the charging and discharging process of dispatchable EVs to participate in the energy market and obtain operational profits. The objective function of the EV aggregator includes the following aspects:

- a) *Energy costs*: The EV aggregator needs to formulate charging plans for EVs based on market electricity prices to assist with EV charging. After determining the charging plan, the aggregator needs to purchase the corresponding amount of electricity based on the plan's demand. Therefore, the electricity purchase cost for EV aggregators should be expressed as follows:

$$F_1 = \sum_{t=1}^T \rho_{cm}(t) (P_{cs}^{dis}(t) + P_{cs}^{undis}(t)) \Delta t \quad (4)$$

where

- $P_{cs}^{dis}(t)$ is the planned charging power for the dispatchable EVs
- $P_{cs}^{undis}(t)$ is the planned charging power for the undispachable EVs
- Δt is the time interval of the energy market
- T is the total number of time periods of the energy market

- b) *Energy service revenue*: All the dispatchable EVs should respond to the charging incentive mechanism and the discharging incentive mechanism, thereby obtaining revenues for providing energy service. The energy service revenue for EV aggregators should be expressed as follows:

$$F_2 = -\sum_{t=1}^T \rho_{ds}(t) P_{ds}(t) \Delta t - \sum_{t=1}^T (1 - \alpha(t)) \rho_{cm}(t) P(t) \Delta t + \sum_{t=1}^T \rho_{cs}(t) P(t) \Delta t \\ + \sum_{t=1}^T \rho_{cm}(t) P_{cs}^{undis}(t) \Delta t + \sum_{t=1}^T \rho_{rt}(t) (P_{ds}(t) + P_{cs}(t)) \Delta t \quad (5)$$

where

- $\sum_{t=1}^T \rho_{ds}(t) P_{ds}(t) \Delta t$ is the total cost for the discharge of the dispatchable EVs,
- $P_{ds}(t)$ is the discharging power of the EV aggregator at time period t
- $\sum_{t=1}^T (1 - \alpha(t)) \rho_{cm}(t) P(t) \Delta t$ is the total cost for charging the dispatchable EVs
- $\sum_{t=1}^T \rho_{cs}(t) P(t) \Delta t$ is the total revenue for charging the dispatchable EVs
- $\sum_{t=1}^T \rho_{cm}(t) P_{cs}^{undis}(t) \Delta t$ is the total revenue for charging the undispachable EVs
- $\sum_{t=1}^T \rho_{rt}(t) (P_{ds}(t) + P_{cs}(t)) \Delta t$ is the total revenue for responding to the charging incentive mechanism and the discharging incentive mechanism
- $\rho_{rt}(t)$ is the subsidized electricity price from the energy market
- $P(t)$ is the net power of the EV aggregator at time period t

$P(t)$ should be calculated as follows:

$$P(t) = P_{cs}^{dis}(t) - P_{ds}(t) + P_{cs}(t) \quad (6)$$

where

$P_{cs}(t)$ is the charging power that exceeds the charging plan

Taking the total operational revenue obtained by the EV aggregator participating in the energy market as the optimization objective, the objective function should be expressed as follows:

$$\max F = -F_1 + F_2 \quad (7)$$

6.1.4 Constraints of the EV aggregator

The EV aggregator adjusts the power response to meet the demand of the energy market; therefore, the net power of EV aggregators $P(t)$ during each time period should meet the following constraints (G. Le et al. [B43]):

$$P^-(t) \leq P(t) \leq P^+(t) \quad (8)$$

$$E^+(t) = \sum_{i=1}^{N_i} s_i^+(t), E^-(t) = \sum_{i=1}^{N_i} s_i^-(t) \quad (9)$$

$$P^+(t) = \min \left\{ \sum_{i=1}^{N_i} \zeta_i(t) p_i^+(t), \frac{E^+(t + \Delta t) - E^+(t)}{\Delta t} \right\} \quad (10)$$

$$P^-(t) = \max \left\{ \sum_{i=1}^{N_i} (1 - \zeta_i(t)) p_i^-(t), \frac{E^-(t + \Delta t) - E^-(t)}{\Delta t} \right\} \quad (11)$$

where

$s_i^+(t)$ is the maximum SoC of EV i at time period t
 $s_i^-(t)$ is the minimum SoC of EV i at time period t
 N_i is the total number of the dispatchable EVs
 $p_i^+(t)$ is the maximum charging power of EV i at time period t
 $p_i^-(t)$ is the maximum discharging power of EV i at time period t

In the above, $\zeta_i(t)$ is a binary variable; $\zeta_i(t) = 1$ denotes that EV i is charging, otherwise, EV i is discharging at time period t . Furthermore, the charging plan of all the EVs in the EV cluster should be satisfied at the end of time period $t = T$.

6.2 National policy recommendations for V2G

6.2.1 Provide technical support and R&D funding

V2G technology requires continuous innovation to optimize its integration into existing energy systems, enhance efficiency, and encourage widespread adoption. For example, technical support and research and development (R&D) funding should be developed in the following aspects:

- a) *Bidirectional energy flow*: Overcoming the bidirectional energy flow is a critical aspect of advancing V2G technology (J. Fang et al. [B25]). The chargers need to be compatible with various EV models and different battery chemistries, and the energy losses should be reduced during conversion processes. Furthermore, developing cost-effective technologies that should be widely deployed without significantly raising the overall cost of EVs or infrastructure is also challenging. Solutions for developing bidirectional charging technology should be explained as follows:

- 1) *Universal standards*: Collaborate with stakeholders such as international standard organizations, automakers, and charging equipment manufacturers to create and adopt universal standards for connectors, communication protocols, and charging processes. Implement rigorous testing and certification processes to help ensure all equipment meets the established standards for safety and interoperability across all vehicle types.
- 2) *Advanced power electronics*: Invest in the development of power electronics, such as silicon carbide and gallium nitride technologies, which offer higher efficiency and lower losses compared to traditional silicon-based systems. Enhance the synergy between power electronics in V2G systems and smart grid infrastructure for dynamic energy management (Mwasilu et al. [B57]).
- b) *Grid integration and stability*: A primary concern is helping to ensure that the energy returned to the grid by EVs does not destabilize the system, especially during periods of peak demand (M. Inci et al. [B36]). The variability in energy supply, as EVs connect and disconnect, requires advanced methods for balancing supply and demand dynamically. This necessitates sophisticated grid management solutions capable of predicting and adjusting to real-time changes. Without these measures, the integration of V2G systems could strain existing grid infrastructure, complicating efforts to maintain consistent power quality and reliability across the network. Solutions for helping to ensure grid integration and stability should be explained as follows:
 - 1) *Smart grid technologies*: Implement smart grid technologies that allow utilities to monitor energy flows in real-time and adjust supply and demand dynamically. Deploy decentralized control systems that enable localized decision-making and energy management, reducing bottlenecks and enhancing grid resilience.
 - 2) *Real-time data analytics*: Develop and deploy algorithms capable of processing large volumes of data to predict usage patterns and optimize energy distribution. Utilize machine learning models to improve prediction accuracy and adapt to changing conditions.
 - 3) *Aggregation services*: Platform-based approaches that aggregate the energy storage capacity of multiple EVs to provide grid services such as load balancing, frequency regulation, and peak shaving. Implement dynamic pricing models that promote energy discharge during high-demand periods while helping to ensure consumer benefit through financial incentives.

6.2.2 Establish distribution pricing policies

V2G technology allows EVs to not only draw power from the grid but also send it back. This bidirectional energy flow introduces a new and flexible energy resource that should be leveraged to stabilize the grid, particularly during peak demand periods. As a result, pricing policies shall be adapted to accommodate and incentivize this new capability. Effective pricing policies should encourage EV owners to participate in V2G programs, offering utilities a distributed method for managing energy loads and reducing the need for additional infrastructure investments. Establishing distribution pricing policies in the context of V2G technology involves creating flexible pricing structures that reflect the new capabilities and dynamics introduced by V2G systems, which should be explained in the following aspects:

- *Time-of-use (ToU) rates*: Higher electricity rates during peak demand times when grid stress is highest and these are typically aligned with morning and evening periods when residential and commercial demand spikes. Reduced rates during times of low demand, usually overnight and mid-afternoon, designed to encourage energy use during these periods (L. Zhao, Z. Yang, and W. J. Lee [B96]). ToU rates should vary seasonally to account for changes in demand, such as higher demands during winter heating or summer cooling periods.
- *Real-time market pricing (RTP)*: Prices are recalibrated every hour (or more frequently) based on wholesale market prices and current grid demand. Using digital platforms to notify users of price changes, encouraging them to adjust usage or supply energy back to the grid (T. Cortes-Arcos et al. [B16]).

The implementation of pricing policies should focus on the following aspects:

- *Technology requirements:* Deployment of advanced metering infrastructure that supports two-way communication between consumers and utilities. Development of mobile apps and online portals that display real-time pricing information to consumers.
- *Regulatory framework:* Approvals from regulatory bodies to establish ToU and RTP frameworks. Consumer protection mechanisms to safeguard against bill shocks, such as price caps during extreme conditions.
- *Consumer engagement:* Educational campaigns to inform consumers about pricing structures and how to optimize their energy usage. Incentives for using automation technologies (like smart thermostats; D. P. Chassin et al. [B13]) to adjust energy usage in response to real-time pricing signals.

6.2.3 Establishment of regulatory bodies

V2G technology blurs the lines between energy producers and consumers, creating a new class of prosumers who should both consume and supply energy. This dynamic demands new regulatory frameworks to help ensure fair market access, grid reliability, and consumer protection. To help ensure the safe and efficient integration of V2G into the existing energy ecosystem, regulatory bodies shall evolve to address issues in the following aspects:

- a) *Pricing mechanisms:* Create a formal approval process for new pricing models, such as ToU and RTP, helping to ensure they align with policy goals and market needs. Allow for dynamic adjustments to pricing models based on grid conditions, market fluctuations, and technological advances (Roozbehani, Dahleh, and Mitter [B69]).
- b) *Consumer protections:* Mandate that all pricing structures and billing practices are transparent and easily understandable to consumers. Implement safeguards to help prevent unfair pricing practices, helping to ensure that all consumers have equitable access to V2G programs and benefits.
- c) *Market fairness:* Helping to ensure that pricing policies allow equitable access to V2G services across different consumer groups, helping to prevent discrimination based on geography, socio-economic status, or other factors. Develop standards for compensating consumers for energy services provided via V2G, reflecting the true value of energy supplied back to the grid (Karfopoulos, Panourgias, and Hatziaargyriou [B41]).
- d) *Monitoring and compliance:* Develop enforcement mechanisms to help ensure compliance with approved pricing policies, including penalties for non-compliance (G. Raman et al. [B66]). Conduct regular reviews and assessments of pricing mechanisms to help ensure alignment with evolving market dynamics and technological advancements.

6.2.4 Enhancing data security and privacy protection

Enhancing data security and privacy protection in the context of V2G technology is crucial to safeguard sensitive information and maintain system integrity (Pokhrel and Hossain [B63]). The measures and strategies for enhancing data security and privacy protection evolve the following aspects:

- a) *Advanced encryption protocols:* Implement strong encryption protocols, such as (B. Schneier et al. [B73]), for data in transit and at rest to help prevent unauthorized access to sensitive information. Use secure communication channels like TLS/SSL (Kumar et al. [B42]) for all data exchanges between vehicles, charging stations, and grid operators.
- b) *Identity and access management:* Require multi-factor authentication for users accessing V2G systems to help ensure that they are only authorized individuals should execute transactions (Nikravan and Reza [B60] and Kumar et al. [B42]). Establish role-based access control to limit access to data and system functionalities based on user roles and responsibilities.

- c) *Data Anonymization and Minimization*: Use techniques such as tokenization and data masking (Roy, Shovon, and Whaiduzzaman [B70] and Badgujar [B7]) to help ensure that personal data cannot be traced back to individuals. Collect only the minimal amount of data necessary for V2G operations, reducing the risk of data breaches.
- d) *Regular security audits and vulnerability assessments*: Conduct regular security audits and penetration testing to identify vulnerabilities and help ensure compliance with security standards. Implement a robust patch management process to rapidly address security vulnerabilities and software updates (Dey, Lahiri, and Zhang [B19]).
- e) *Incident response and recovery plans*: Develop and maintain a detailed incident response plan to quickly detect, respond to, and mitigate any breaches or attacks. Conduct regular drills and simulations to help ensure that response teams are prepared to handle actual security incidents (Novak and Ivanov [B61]).

6.3 V2G social impact

6.3.1 General

According to V2G-Hub, there are currently 151 V2G cases, including 105 in Europe, 23 in the United States, and 15 in Asia. The pilot projects include time shifting (peak shaving and valley filling), frequency modulation, distribution network congestion management, emergency backup, peak-valley arbitrage, and spinning reserve. This recommended practice provides three use cases for V2G pilots in China, the UK, and the United States, which indicates the social impact of V2G technologies in peak shaving and valley filling, reducing CO₂ emissions, and energy support for the power grid.

6.3.2 Use case 1—Demand response for different charging facilities

In 2019, Shanghai, China, carried out three pilot activities for EVs participating in demand response. The specific pilot times are June 7, 2019, from 2 a.m. to 5 a.m., August 9, 2019, from 12 p.m. to 2 p.m., and December 5, 2019, from 10 a.m. to 11 a.m. The specific route, requirements, and compensation coefficient rules should refer to (Natural Resources Defense Council, 2020 [B58]).

The number of private charging piles participating in the demand response on June 7, 2019, was 159. The average charging load reached 1068 kW, 7.8 times the charging load for the same period on regular days. In addition, on the response day of June 7, the average charging amount for 159 response charging piles was 64.5 kWh, significantly higher than the daily average charging amount of 9.35 kWh on regular days (June 1). This demonstrates that price incentives have a strong guiding effect on electric vehicle owners.

The number of dedicated charging piles participating in the demand response on August 9, 2019, was 110. The average charging power was 43 kW, a decrease of 75% compared to the average charging power of 167 kW on regular days. Compared to private charging piles, it shows better participation, facilitating the management of centralized charging and discharging behaviors.

6.3.3 Use case 2—Reducing carbon emissions

From September 2018 to January 2022, with a grant of £3.77 million, a project in the UK aimed to demonstrate V2G in various real-world situations to gain technical, customer, and commercial insights into this emerging technology (“EV-elocity Project Final Report” [B23]).

An investigation by a team from a university in UK, funded by the project, aimed to propose methods to reduce CO₂ emissions using V2G. To achieve this, the researchers integrated data sets on vehicle use behavior, building energy demand, local renewable energy generation, and grid carbon intensity. This analysis included creating a strategy for charging and discharging the vehicles that considered these variables.

As shown in Waldron et al. [B83], V2G reduced the average carbon intensity per kilowatt from 223.8 gCO₂/kWh (with a baseline demand of the building without electric vehicles) or 224 gCO₂/kWh (with unmanaged charging) to 218.9 gCO₂/kWh using V2G. This indicates that V2G was able to shift energy demand to periods when renewable energy generation was high and the grid carbon intensity was low, without impacting current service provision.

6.3.4 Use case 3—V2G service platform

A school in California is the first in the United States to adopt V2G technology and achieve full electrification of school buses. Their student transportation service platform provides a fleet consisting of 74 electric school buses and bidirectional charging equipment for this school district. The utility company provides a 2.7 MW load for its charging and discharging facilities, and the fleet is managed by AI-powered technology platform. The V2G pilot of this fleet has shown significant results, achieving zero emissions and returning 2.1 GWh of energy to the grid annually as a virtual power plant, which is significant for community carbon reduction and is expected to reduce greenhouse gas emissions by 25 000 tons. The platform has set a goal of electrifying 10 000 bidirectional school buses and plans to deploy a 100% electric fleet nationwide in the coming years (Natural Resources Defense Council, 2024 [B59]).

Annex A

(informative)

Use case—V2G-based hierarchical partitioned aggregation

A.1 Overview

The V2G-based hierarchical partitioned aggregation model enhances electric vehicle (EV) integration by structuring communication and control into local, regional, and central levels. This approach optimizes energy management, helps ensure scalability, and facilitates the provision of ancillary grid services. The following use cases illustrate practical applications of this model in diverse scenarios.

A.2 Use case 1—Urban peak load management

A.2.1 Scenario

A metropolitan area with high EV penetration experiences frequent peak demand periods, stressing the grid, and the optimal load matching performance should be achieved in a net-zero energy city with the V2G aggregation scheme (Fachrizal et al.[\[B24\]](#)).

A.2.2 Implementation

Implementation is as follows:

- *Local level:* EVs in residential neighborhoods and commercial districts are grouped into local clusters. Aggregators manage charging schedules based on local grid load and renewable energy availability.
- *Regional level:* Regional control centers consolidate data from local clusters and coordinate demand response strategies across neighborhoods.
- *Central level:* A central system oversees inter-regional coordination, deploying surplus energy from low-demand areas to high-demand zones.

A.2.3 Benefits

Benefits are as follows:

- Reduces peak load stress through distributed load balancing
- Aligns EV charging with renewable energy generation, reducing emissions

A.3 Use case 2—Renewable energy integration in rural areas

A.3.1 Scenario

A rural area with abundant solar and wind resources faces challenges in utilizing excess renewable energy during low-demand periods.

A.3.2 Implementation

Implementation is as follows:

- *Local level:* EVs near renewable energy generation sites are grouped into clusters. Aggregators schedule charging during periods of excess generation.
- *Regional level:* Regional centers manage energy distribution across clusters, helping to ensure that renewable energy is utilized efficiently.
- *Central level:* A central system monitors renewable energy output and adjusts regional charging strategies to reduce curtailment.

A.3.3 Benefits

Benefits are as follows:

- Enhances renewable energy utilization
- Provides ancillary services, such as frequency regulation, through V2G-enabled discharging

A.4 Use case 3—Disaster recovery and emergency management

A.4.1 Scenario

A natural disaster causes localized power outages, disrupting grid operations in affected areas. In January 2011, a local government area in the West Moreton region of South East Queensland experienced catastrophic flooding, resulting in significant loss of life, property damage, and prolonged power outages. The recovery process required substantial resources and coordination (Medford [B50]).

A.4.2 Emergency power supply for shelters and medical facilities

The emergency power supply for shelters and medical facilities is as follows:

- a) Implementation:
 - 1) Equip emergency response vehicles and EVs owned by NGOs with V2G capability.
 - 2) Establish V2G-enabled charging stations at key locations, such as evacuation shelters and medical facilities.
- b) *Example:* V2G-enabled EVs could have provided temporary power to shelters and medical facilities, helping to ensure *continuous* operation of essential services like lighting, communication, and medical equipment. EVs equipped with V2G should serve as mobile power sources, supplying electricity to critical infrastructure during outages.

A.4.3 Powering communication networks

Powering communication networks is as follows:

- a) *Implementation:* Deploy V2G-enabled EVs to power critical communication infrastructure, such as mobile towers and emergency communication hubs.

- b) *Example:* Using V2G technology, EVs could have powered communication networks, allowing for better coordination of relief efforts and helping to ensure residents stayed informed and connected with rescue teams.

A.4.4 Supporting search and rescue operations

Supporting search and rescue operations is as follows:

- a) *Implementation:* Use V2G-enabled EVs to power search and rescue equipment, such as lighting, water pumps, and cutting tools.
- b) *Example:* V2G-enabled EVs could have powered portable floodlights for night-time search and rescue operations, enhancing the effectiveness and safety of the rescue teams.

A.5 Use case 4—Market participation in energy trading

A.5.1 Scenario

In a pilot project led by a regional research institute and a municipal utility operator, EV fleets were aggregated to participate in energy markets by offering frequency-containment services. Vehicles not in use during nighttime hours were scheduled for discharge, generating quantifiable revenue and demonstrating the commercial viability of V2G-enabled trading (“Vehicle-to-Grid Charging for Electric Cars” [B82]).

A.5.2 Implementation

Implementation is as follows:

- *Local level:* EV clusters are managed by local aggregators to optimize charging and discharging schedules for market participation.
- *Regional level:* Regional centers aggregate data from local clusters to create bids for energy markets, increasing collective revenue.
- *Central level:* The central management system oversees market trends and dynamically adjusts regional participation strategies to align with grid requirements.

A.5.3 Benefits

Benefits are as follows:

- Generates revenue for EV owners
- Stabilizes energy prices through demand-side flexibility

A.6 Use case 5—Fleet management for public transportation

A.6.1 Scenario

A city operates a fleet of electric buses and taxis, requiring efficient energy management for daily operations. This initiative of a project in California (Schorske et al. [B75]) examined the feasibility of integrating electric buses with the grid to provide V2G services. The project highlighted the need for clear regulatory guidelines to effectively utilize electric fleets for augmenting stationary battery storage capacities.

A.6.2 Implementation

Implementation is as follows:

- *Local level:* Each depot forms a local cluster where charging schedules are optimized based on route demands and vehicle availability.
- *Regional level:* Regional centers coordinate depots to balance energy demand and prioritize charging for high-utilization routes.
- *Central level:* A central system monitors fleet performance, helping to ensure system-wide efficiency and reliability.

A.6.3 Benefits

Benefits are as follows:

- Reduce operational costs for public transportation fleets
- Improve service reliability through proactive energy management

A.7 Use case 6—A network interactive comprehensive demonstration station in China

A.7.1 Scenario

A car network interactive comprehensive demonstration station in Jiangxi Province, China, integrates PV, storage, charging, discharging, battery swapping, and inspection, with a total area of 5818 m². It has completed the installation of 53 charging and swapping facilities, including 1 super liquid-cooled charging pile of 640 kW, 5 bidirectional V2G charging and discharging piles of 120 kW, and 12 high-power charging piles of 160 kW and above. It has also installed a photovoltaic canopy with a power generation capacity of 140 kW and 80 kWh of energy storage.

A.7.2 Benefits

The car network interactive comprehensive demonstration station builds a site-level load resource library, integrating various loads such as photovoltaics, energy storage, charging, and discharging. V2G demonstrations in various scenarios, such as passenger cars and battery swap stations, have a discharge capacity of up to 1800 kW. Through “Smart Energy Saving + V2G,” the demonstration application of the entire V2G process has been realized.

A.8 Use case 7—Operation fleet charging—Power utilization optimization

A.8.1 Scenario

A commercial charging facility operated by a local utility in southern China integrates V2G functionality, photovoltaics, and stationary storage. The site connects to both spot electricity markets and demand response programs, enabling pricing-driven fleet charging optimization. See [Figure A.1](#).

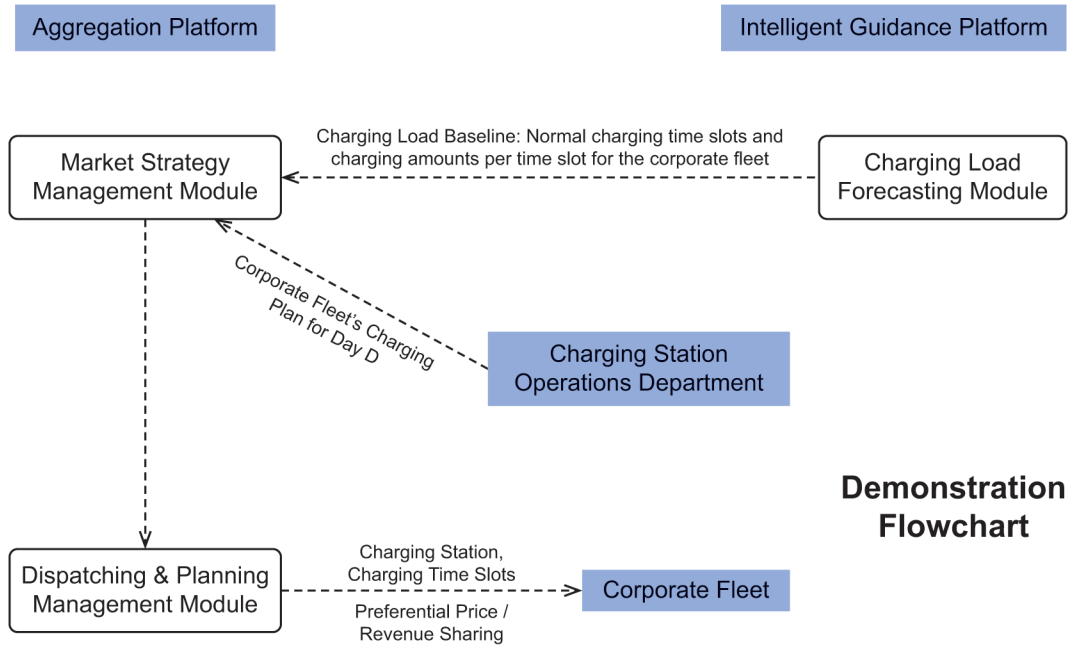


Figure A.1—The station-level control framework

A.8.2 Benefits

This integration achieved a 30.84% revenue increase over traditional charging models, generating ¥1827 (approximately US\$252) in a one-week spot-price response pilot, while reducing operational fleet costs by 15% to 20% through price-guided charging shifts and improving nighttime wind-power consumption efficiency via coordinated grid response.

Annex B

(informative)

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